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MORPHOLOGICAL PROCESSING DURING VISUAL WORD RECOGNITION: MECHANISMS OF DYNAMICS AND ACQUISITION

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CHAPTER 5

A Variation on the Orthography-Semantics Consistency Metric

The impact of morphemes on word processing is due at least in part to the role they play in informing a word's meaning: *dict* is not only helpful for recognizing *dictate* because of its frequency across contexts, but also because of the information it provides about the meaning of *dictate*. Thus, it would presumably be informative to keep track of how useful a given morpheme is as a clue to the word's meaning. Put another way, how semantically similar are words that contain that morpheme? Modern computational linguistics methods for obtaining distributed semantic vector representations of words (e.g., Landauer & Dumais, 1997; Pennington et al., 2014) have made it more feasible to approximate a morpheme's semantic utility (see Amenta et al., 2020b). The recently developed *orthography-semantics consistency* metric takes advantage of this.

Introduced by Marelli et al. (2015) and discussed further by Marelli & Amenta (2018) and (Amenta et al., 2020a), orthography-semantics consistency (OSC) refers to the frequency-weighted semantic similarity between a standalone stem and the words containing it. Stems that are overall closer in meaning to words containing them, especially high-frequency words containing them, will have a higher consistency value. For example, words that contain the letter sequence *trust* tend to be fairly similar in meaning to *trust* (*distrust*, *trustworthy*, *trustee*), while words that contain *whisk* are not very similar to *whisk* (*whisking*, *whisker*, *whiskey*). Thus, *trust* has high OSC while *whisk* has low OSC.

As mentioned in Chapter 2, the original purpose of calculating this consistency metric was to quantify a potential confound present in priming studies of semantic transparency. Marelli et al. (2015) demonstrated that the targets used in several previous masked priming studies of semantic transparency differ significantly in OSC across conditions, and they argued that this difference could be driving commonly observed faster response times to transparent targets than to opaque targets when preceded by unrelated primes. In other words, lower-OSC words like WHISK, often used in opaque conditions following WHISKER, are responded to more slowly than higher-OSC transparent targets like CHEER (primed by CHEERFUL), regardless of what prime precedes them. Because the targets in transparent conditions are already being recognized quickly, the degree to which their recognition could be further facilitated is reduced, constraining transparent priming magnitudes to a level more comparable with those of opaque priming. Following this paper, subsequent work on OSC by these authors refined its calculation to better predict lexical decision latencies (Marelli & Amenta, 2018) and investigated its relation to masked priming other than as a potential confound (Amenta et al., 2020a). Very recently, Siegelman et al. (2022) also explored the operational definition of OSC and demonstrated that effects of OSC are more evident in tasks that place more emphasis on word semantics relative to word phonology (i.e., lexical decision but not reading words aloud).

The Marelli & Amenta (2018) metric of a morpheme's OSC is calculated by identifying all of the words containing the stem (regardless of whether they're considered morphologically related to the stem), and calculating the frequency-weighted average cosine similarity between the semantics of each such word and that of the standalone stem. More formally:

$$\text{OSC}_s(t) = \frac{\sum_{x=1}^{k} \cos(\mathbf{r}_t, \mathbf{r}_x) * f_x}{\sum_{x=1}^{k} f_x} \tag{5.1}$$

In this equation, $\mathbf{r}_t$ is a semantic vector for the target stem, $\mathbf{r}_x$ is a semantic vector for each word x containing the target stem (i.e., each "orthographic neighbor"), and f_x is the frequency of each word containing the target stem. Semantic vectors are obtained using distributional semantics techniques, which will be described in more detail in subsequent sections.

Marelli & Amenta (2018) demonstrated that their OSC metric predicts lexical decision times from the English Lexicon Project (Balota et al., 2007), after controlling for word frequency, orthographic length, and morphological family size. OSC also predicts the magnitude of facilitation effected by morphological masked priming (Amenta et al., 2020a), suggesting that the degree to which morphologically related words boost each others' recognition is in part determined by the OSC of the shared morpheme.

However, as a measure of a morpheme's semantic consistency across the words in which it appears, the OSC metric proposed by Marelli & Amenta (2018) has two notable shortcomings. First, distributed semantics models typically only provide semantic vectors for letter strings that exist as whole words. Consequently, by calculating all semantic distances relative to the standalone stem's meaning, OSC cannot be calculated for morphemes that do not exist as independent words (i.e., bound stems like *dict* and *cept*, and affixes like *pre* and *ness*). Second, using the standalone stem as the semantic distance reference point gives it a stronger role in determining OSC, relying on the assumption that the standalone stem reserves a unique status in contributing to a morpheme's semantic significance among words containing the string. While this may be true, we do not know of any evidence to suggest that it is, and furthermore there are at least some cases where the meaning of the stem word is aberrant from meanings of the rest of the words containing it (e.g., *man* can mean "male human" or "work or operate", as in "man the pumps", but the majority of words containing the word *man* relate primarily to the first meaning). Using the stem as the reference point to which all longer words are compared in such cases would result in a skewed metric of how informative that letter string is for the set of words not including the stem.

The current work proposes and evaluates a variation on the OSC metric. In this variation, all words containing the letter string of interest contribute to the calculation proportional to their frequency. Adapting how OSC is quantified to allow its computation for non-word letter strings advances the primary goals of this dissertation by expanding the implications and utility of this metric. The ability to calculate OSC for any string of letters is key to determining whether morphological sensitivity is driven by the utility of certain letter strings for cuing word meanings, as is most explicitly predicted by the distributed account of morphological processing. A refined OSC metric also facilitates adopting a graded view of morphology in psycholinguistics (at least in languages with concatenative morphology), as letter strings can be localized along a spectrum of semantic utility instead of being categorized as "morpheme" or "non-morpheme".

5.1 METHODS

The primary innovation for this variation of OSC is the reference point to which semantic distance was calculated. Instead of calculating semantic distances from all words containing the letter string of interest (the stem's "orthographic neighbors": WHISKY, WHISKER, etc.) to the target on its own (WHISK), a frequency-weighted centroid for the cluster of meanings of words containing the string (shown as an "X" in Figure 5.1) is used as the distance reference. Consistency is then calculated as the mean cosine similarity between semantic vectors of words containing the string and this centroid. In formal terms,

$$\mathrm{OSC}_c(t) = \mathbb{E}_{x=1}^{n}\big[\cos(\mathbf{r_x}, \mathbf{c})\big] \tag{5.2}$$

In this case, $\mathbf{r_x}$ is a semantic vector for each word containing the target stem, including the standalone stem itself, if it exists. The variable $\mathbf{c}$ is the frequency-weighted centroid:

$$\mathbf{c} = \frac{\sum_{j=1}^{n} \mathbf{r_j} * f_j}{\sum_{j=1}^{n} f_j} \tag{5.3}$$

This variation of OSC is a metric of the tightness of the cluster of word meanings, or the semantic similarity of all words containing the letter string of interest. The letter string itself does not need to exist as a standalone word in order for OSC to be calculated, and the set of words containing the letter string is assessed holistically, not relative to a single word.

Although the most noteworthy change in the new OSC metric is the reference point for semantic distances, different frequencies and semantic vectors were also used. Instead

Figure 5.1: Reference points for calculating semantic relatedness via the original OSC metric (red +) and the new proposed variation (red X). Semantic space reduced to two dimensions for visualization using multidimensional scaling. Point size corresponds to word frequency.

	OSC_c	OSC_s	OSC_o
OSC_c	1.00	0.76	0.62
OSC_s	-	1.00	0.55
OSC_o	-	-	1.00

Table 5.1: Correlation matrix for the three OSC metrics.

of constructing original semantic spaces, as done by Marelli & Amenta (2018), the GLoVe 300-dimensional semantic space was used (Pennington et al., 2014). Frequency values based on the 17.9 billion word COBUILD English corpus were obtained from the CELEX database (Baayen et al., 1995). Additionally, words that contained the letter string of interest were identified from the set of words present in both the CELEX database and the SUBTLEX-US database (Brysbaert & New, 2009). Because of these differences, the new cluster-centered metric (OSC_c) was compared to a metric generated by applying the stem-centered calculation to CELEX frequencies and GLoVe representations (OSC_s), as well as to the original OSC values (OSC_o) from Marelli & Amenta (2018). Correlations between the three OSC metrics are shown in Table 5.1.

Model	OSC Effect	Variance Explained	AIC	BIC
Baseline		0.41	-30370.91	-30335.46
OSC_c	-0.0029	0.41	-30369.74	-30327.21
OSC_s	-0.015***	0.42	-30412.08	-30369.55
OSC_o	-0.011***	0.42	-30403.65	-30361.12

Table 5.2: Comparing OSC metrics as predictors of English Lexicon Project reaction times. Effects with *** had p-values less than 0.0001. OSC_c was not a significant predictor.

5.2 RESULTS

Analyses were conducted in R (version 4.0.3; R Core Team, 2017), and linear mixed-effects models were fit using the *lme4* package (version 1.1-26; Bates et al., 2015). The *lmerTest* package was used to obtain p-values for mixed effects terms (version 3.1-3; Kuznetsova et al., 2017).

The three OSC metrics were first assessed in terms of how well they captured variation in lexical decision reaction times. These analyses followed the comparisons described by Marelli & Amenta (2018), but using words and reaction times from the English Lexicon Project (Balota et al., 2007) instead of the British Lexicon Project (Keuleers et al., 2012). Similar to Marelli & Amenta (2018), OSC values were assessed relative to a baseline model including frequency (from the SUBTLEX-US database; Brysbaert & New, 2009), morphological family size (from CELEX; Baayen et al., 1995), and orthographic length as predictors. Words that were not found to be embedded in any other words were excluded, as were words not present in the GloVe database, and those for which family size or SUBTLEX-US frequency could not be obtained. This left 8,857 items for analysis.

Results are summarized in Table 5.2. The model that best explained variance in lexical decision reaction times was the one including the new stem-centered OSC metric, followed by the original OSC from Marelli & Amenta (2018). Variance explained did not differ dramatically across the models. The cluster-centered variation increased AIC and

BIC values when added to the baseline model (indicating a worse fit), and did not significantly predict variations in reaction times beyond baseline predictors. Interestingly, the cluster-centered and stem-centered metrics explained variance in the other model's residuals (for OSC$_c$: $t = 0.0043$,$p = 0.06$; for OSC$_t$: $t = -0.0087$, $p < 0.001$), suggesting they capture different aspects of variance in reaction times.

The next analyses concerned data from previous studies of masked morphological priming, and followed procedures reported by Amenta et al. (2020a). Mean reaction times for each prime-target pair were taken from Rastle et al. (2000), Rastle et al. (2004), Andrews & Lo (2013), and Jared et al. (2017). Prime-target pairs for which the target OSC could not be calculated or for which frequencies, orthographic neighborhoods or family sizes could not be obtained were excluded, leaving average priming magnitudes for 676 related prime-target pairs (535 unique pairs, 513 unique targets). These pairs were categorized as orthographic, opaque, quasitransparent or transparent, corresponding to how they were categorized in the study from which they were taken.

Four models were fit to the priming data. All used mean priming magnitude (reaction times for unrelated minus related pairs) as a dependent variable and included target length, log-transformed morphological family size, log-transformed prime and target frequency, orthographic neighborhood (OLD20, obtained from Balota et al., 2007) and transparency (cosine similarity of the prime and target GloVe representations) as covariates. Interaction terms for OSC and the two frequency covariates, and random effects for target and study, were also included.

The OSC$_o$ model showed a marginal main effect of OSC ($t = 21.19, p = 0.08$), and a significant positive interaction with prime frequency ($t = 19.56, p = 0.04$). The OSC$_s$ model also yielded a significant positive interaction of OSC and prime frequency ($t = 37.51, p = 0.005$), although no main effect of OSC was found ($t = 6.37, p > 0.5$). The OSC$_c$ model yielded a strong, positive main effect on priming magnitudes ($t = 72.60, p = 0.004$) and, though lacking a positive interaction with prime frequency ($t = 7.16, p > 0.5$), showed a marginal negative interaction with target frequency ($t = -27.44, p = 0.09$). The OSC$_c$ model had the lowest AIC and BIC values (6879.81 and 6938.52, respectively), very closely followed by OSC$_s$ (6880.85 and 6939.56), OSC$_o$ (6881.04 and 6939.75), and the baseline model (6907.40 and 6952.56). Similar patterns of effects were observed when prime-target pairs assigned to the orthographic condition were excluded.

5.3 DISCUSSION

The cluster-centered OSC metric is a worse predictor of unprimed lexical decision RTs, but is as good or perhaps slightly better at explaining variance in priming magnitudes, relative to the two stem-centered OSCs. That the cluster-centered OSC as a poorer predictor of unprimed lexical decision makes sense, given its reduced focus on the meaning of the standalone stem. When calculating the stem-centered metrics, the meanings of all words containing it are compared to the stem's standalone meaning, but the cluster-centered calculation does not give the standalone stem's meaning a privileged status. Thus, the stem-centered OSC may be a better metric for the *word*, whereas the cluster-centered OSC better captures the consistency of the *letter string* across all its appearances.

Aligning with this interpretation, the main effect of OSC on priming magnitudes was significantly positive for the cluster-centered metric, while being marginal for the original metric from Marelli & Amenta (2018) and nonexistent for the re-calculated stem-centered metric. Taking a distributed view of morphological processing, one might expect that morphological priming magnitudes would be stronger for prime-target pairs with a more semantically consistent shared morpheme. This prediction follows from the fact that more consistent morphemes would be more useful entities to leverage in the process of mapping from orthography to semantics. Thus, the observation of a stronger main effect for the cluster-centered metric on priming magnitudes suggests that it better captures the letter string's consistency across its appearances (the letter string -TRUST- across TRUSTING, UNTRUSTWORTHY, DISTRUST), rather than that of the *word* (TRUST).

The cluster-centered OSC model showed a marginal negative interaction of OSC with target frequency, while the stem-centered OSC models yielded a positive interaction with stem frequency. The cause for this difference in observed interactions can be explained at least partially by the word-metric versus morpheme-metric distinction just discussed. If the stem-centered metric is a better measure of the word's consistency than of the morpheme's consistency, then the effect of this metric on speed of target recognition would be stronger, and the additional effect of the metric on priming magnitude would only be detectable when the prime is also high-frequency. Conversely, the negative interaction of the cluster-centered OSC metric with target frequency would suggest that the facilitating effect of OSC is stronger when the target is low-frequency, because high-frequency targets can only be recognized slightly more quickly than they already are and thus can only be facilitated to a small degree by their relatedness to the prime. Put more simply, an interaction with prime characteristics makes more sense if the predictor best describes the target word, while an interaction with target characteristics is expected if the predictor best describes the relationship between prime and target.

5.4 CONCLUSION

The revised metric of orthography-semantics consistency proposed in this chapter appears to better capture the consistency of a particular string of letters across its contexts (TRUSTING, UNTRUSTWORTHY, etc.), without giving undue priority to the meaning of the standalone word (TRUST). Additionally, the new approach to calculation can be applied to any letter string, not being limited to letter strings that exist as independent words. These improvements in how OSC is operationalized will enhance our ability to investigate morphological processes during visual word recognition, and their acquisition, in the context of a graded spectrum of semantic utility from non-morpheme to morpheme. The experiments conducted in Chapters 6 and 8 use this variation on OSC to inform stimuli selection and conduct statistical analyses.

CHAPTER 6

Effects of Orthography-Semantics Consistency on Morphological Processing in Skilled Adult Readers

In skilled adult readers, morphological processing mechanisms have been shaped by pressures to recognize the written words they are exposed to quickly and accurately. This learning context means that the brain must begin extracting and making use of information as soon as possible, without waiting until the identity of the presented word is certain. As discussed in Chapter 2, the masked primed lexical decision paradigm provides insight into what factors most strongly influence word perception before it is recognized, by presenting the prime word very briefly (about 50 milliseconds). In a masked priming context, the factors that appear to impact morphological priming magnitudes have more to do with the word's appearance, and the general roles of the word and its substrings in the language more broadly, as this is the type of information most notably represented when the target appears. A primary example of this is semantically opaque morphological priming in English. Rastle et al. (2004) demonstrated that prime-target pairs such as CORNER-CORN and DEPARTMENT-DEPART show significant facilitation in a masked priming context relative to merely orthographically related pairs (CASHEW-CASH; also see Feldman & Soltano, 1999; Marslen-Wilson et al., 2008; Fiorentino & Fund-Reznicek, 2009). As lexical processing continues, factors more particular to the word's whole form, and semantic context, become more relevant. For instance, Rastle et al. (2000) demonstrated that when primes are presented long enough to be consciously processed (e.g., 250 milliseconds), opaque priming effects become negligible or slightly inhibitory, while semantic priming (SKIP-MISS) strengthens and semantically transparent priming remains robustly facilitatory.

Semantically opaque morphological priming, as discussed previously, has become a linchpin result in support of the decomposition perspective (Rastle & Davis, 2008). The pattern of effects described above is clearly compatible with the idea that at some phase of complex word processing, the word is decomposed into morphological constituents prior to recognition or semantic access. However, of the many studies conducted on the topic of opaque morphological priming, few go beyond the question of *whether* it occurs to explore *for which items* it occurs.

From the distributed perspective, opaque priming provides evidence that some morphological paradigms are so consistently significant, the brain starts reacting to their presence to get a head start on lexical retrieval even in cases where their presence is misleading (i.e., the presence of DEPART in DEPARTMENT). From this explanation naturally follows the prediction that opaque morphological priming should be stronger for more semantically consistent stems. Armed with our revised, cluster-centered metric of OSC from Chapter 5, we can begin to test this prediction.

To our knowledge, the only other study that has investigated the effect of OSC on morphological priming magnitudes is Amenta et al. (2020a). They found that OSC interacts with prime frequency to predict morphological priming, such that priming increases with prime frequency for high-OSC stems, while priming decreases with prime frequency for low-OSC stems. Their explanation of this effect is well phrased, and aligns with our interpretation (p. 1553):

> ...in highly consistent families, a frequent prime is a blessing, because form is a reliable cue to meaning and thus primes do yield exploitable information on the target; thus, the stronger the prime, the easier target processing. When form to meaning mapping is instead quite inconsistent in the family, a frequent prime hinders target identification, as form is not a reliable cue to meaning and thus primes are not likely to yield information on the target. (p. 1553)

However, three shortcomings in this study make further investigation necessary. First, analyses used the stem-centered calculation of OSC, and thus are influenced by this metric's bias in favor of the meaning of the stand-alone stem. As described in Chapter 5, analysing the same data used in Amenta et al. (2020a) with our revised OSC metric in yielded a notably different pattern of results, including a significant interaction with target frequency but not prime frequency. Second, the effects described do not differentiate between transparent and opaque priming, and in fact orthographic prime-target pairs were included in this analysis as well. Third, these analyses were conducted with data from prior morphological priming studies; thus, OSC was not maximally varied across stimuli to optimize investigation of its effect on priming, and it was confounded with transparency (OSC was higher among targets of the transparent condition than among targets of the opaque condition; see Marelli et al., 2015). The present study was designed to address these issues with the aim of illuminating the effect of OSC on both semantically opaque and semantically transparent morphological priming.

6.1 STUDY 1: THE EFFECT OF OSC ON MASKED MORPHOLOGICAL PRIMING

This first study looks at the effect of OSC on two types of morphological priming, semantically transparent (GOVERNMENT-GOVERN) and semantically opaque (DEPARTMENT-DEPART), when the prime is masked and briefly presented to prevent recognition. Of primary interest is whether OSC positively predicts opaque masked priming magnitudes. Such an effect would support the distributed perspective: Word recognition processes are most likely to be sensitive to the presence of stems that are generally "semantically useful", and this would be easiest to detect when the meaning of the prime does not also provide semantic information about the target. By including a transparent condition as well, however, we gain an opportunity to observe whether the effect of OSC on morphological priming differs for semantically opaque versus semantically transparent prime-target pairs. Masked priming taps into representations during lexical processing that are less semantically-informed, and thus may allow for the effects of OSC on even transparent priming magnitudes to be detected. Alternatively, the utility of the prime's full word form for recognizing the target may lead to neutral, or even inhibitory effects of stem OSC.

6.1.1 Methods

6.1.1.1 Participants

Participants consisted of 62 undergraduates at a private university in Western Pennsylvania, completing the experiment for credit in a psychology course. All were required to be native speakers of English with normal or corrected-to-normal vision.

6.1.1.2 Stimuli

Prime-target pairs were selected from a set of 278 semantically opaque prime-target pairs and 326 semantically transparent prime-target pairs used in prior studies of morphological priming (Rastle et al., 2000; Beyersmann et al., 2012a; Marslen-Wilson et al., 2008; Jared et al., 2017; Diependaele et al., 2013; Feldman et al., 2009; Beyersmann et al., 2016c; Li et al., 2017; Rueckl & Aicher, 2008; Morris et al., 2007, see description of post-hoc analysis in Chapter 3 for detail). Frequencies, stem family sizes and affix family sizes for all words were obtained via CELEX (Baayen et al., 1995) and orthographic neighborhoods (OLD20) and bigram frequencies were taken from the English Lexicon Project (Balota et al., 2007). Orthography-semantics consistency (OSC_c) values were of the cluster-centered variation described and calculated in Chapter 5.

A publicly available software package for Stochastic Optimization of Stimuli (SOS!, Armstrong et al., 2012) was used to select two 100-item sets: one opaque and one transparent. Items were selected such that OSC_c varied maximally within each set, while remaining minimally correlated with prime and target frequencies, lengths, orthographic neighborhoods and bigram frequencies, as well as stem and affix family sizes, and the number of words in which the stem was found embedded during calculation of OSC. (See Appendix A for the complete set of experimental stimuli used.) Correlations between OSC and these other item metrics were low and non-significant (all rs < 0.2 in magnitude, all ps > 0.05) with the exception of length and orthographic neighborhood in the opaque condition, which were both significantly correlated with OSC (for length: $r = 0.33$, $p < 0.0001$; for OLD20: $r = 0.26$, $p < 0.01$). Length and OLD20 were strongly correlated across the stimulus set ($r = 0.67$, $p < 0.0001$) so we attributed this to the same shortcoming of the opaque set and accounted for this in analysis.

Real word targets were paired with both related primes (pre-determined based on prime-target pairings from prior studies) and unrelated primes (related primes reassigned such that prime and target had distinct consonant onsets and less than half of the letters in the prime were in the target). Each participant saw a given target once, with relatedness of the preceding prime counterbalanced across participants, and performed lexical decision on a total of 400 word and nonword targets.

One hundred thirty-five prime-target pairs (neither prime nor target was complex) were added as filler, to achieve a proportion of 30% related primes for real word targets (following the procedure of Rastle et al., 2000). Targets in this set were between 3 and 5 letters long, while primes were between 5 and 10 letters long, such that relative lengths were comparable to those of the experimental pairs.

Nonwords were obtained for the lexical decision task via three different methods. One hundred were created by permuting the consonant onsets with respect to the remaining strings of the target words until no string combinations created a real word, a near misspelling of a real word (SEAP), a common proper noun (DAVE) or a slang word (RAD). These nonwords were paired with real-word primes that were matched on frequency and length with the experimental primes. One hundred fifty additional nonwords, constrained to be within 3 to 6 letters in length, were taken from the English Lexicon Project (Balota et al., 2007), and paired with real words between 5 and 10 letters in length. Finally, 50 nonwords were extracted as the "stems" of pseudo-complex (e.g., BANT from BANTER or CAPT from CAPTION), and paired with their source word as a prime. Overall, each participant responded to 335 words (200 experimental and 135 filler) and 300 nonwords.

6.1.1.3 *Procedure*

All data collection was done remotely, using jsPsych (De Leeuw, 2015) for experiment presentation, the jspsych-psychophysics plugin (Kuroki, 2021) for masked priming trials presentation [1], and the jspsych-virtual-chinrest plugin to account for participants' screen size in determining stimulus font sizes (Li et al., 2020). Although participants sat at variable distances from the screen on which stimuli were presented, the height of presented text was consistently 0.8 cm; assuming distances between participants' eyes and their screens fell between 30 and 60 cm, this would yield a visual angle of between 0.8 and 1.5 degrees. The study was unmoderated: participants followed the link to the study at their convenience, and read instructions and completed the study in one sitting without experimenter supervision.

Following consent procedures and screen size calibration, participants read instructions that advised them to respond with whether each string of letters was a real word or not, as quickly and accurately as possible. The F and J keys were used for responses, and which key corresponded to "yes" was randomized for each participant. Sixteen practice trials were administered, after which participants received a report of their accuracy and reaction time, given feedback on their statistics (e.g., "Looks like you understand the task! But try to respond more quickly"), and given the option to repeat instructions and practice or begin the experiment. The mean reaction time threshold for recommending faster responses was 720 ms, and participants were advised to review the instructions if their accuracy was below 85%.

The main experiment was broken up by two optional 30-second breaks, such that

[1] The jspsych-psychophysics plugin allows visual stimuli presentations to synchronize with the refresh rate of the display. This is essential for ensuring that masked primes are displayed for the same or as similar as possible amounts of time across the various computers participants are using when they follow the experiment link. See the links at https://osf.io/pj4sb/wiki/home/ under "Histograms related to Table 4" for duration distributions of a stimulus set to last 50 ms across various computers and web browsers, with versus without the jspsych-psychophysics plugin.

participants made continuous lexical decisions for about 10 minutes at a time. The whole experiment lasted 35 minutes on average.

Lexical decision trials began with a fixation cross in the center of the screen for 500 ms, followed by a mask of hash marks (########) for 500 ms. The uppercase prime word was then flashed for 50 ms, and immediately followed by the lowercase target word. The target word was shown for 500 ms or until a response was given, and the trial ended when a response was given, or 2000 ms after the target word first appeared if no response was provided by that time. Participants were informed of the time limit for responses during instructions. A blank screen was shown for one second between trials. This masked priming procedure closely follows the procedure described in Jared et al. (2017).

Following the masked priming task, participants were asked about their experience of the masked priming trials as a check of the effectiveness of this paradigm in a remote context. They were first asked if they noticed anything else beyond the cross, mask, and target being shown; then they were asked to describe what they noticed, how often they noticed it, and finally how often they were able to recognize the uppercase word. Fifty-five percent of participants reported not noticing anything else other than the fixation cross, hash mask, and target word, indicating they were not aware of the 50 ms prime. Of the 25 participants who did detect the flashed stimuli, most only detected that there were letters being shown but were unsure if they formed actual words, and only 1 reported being able to consistently identify the prime word.

6.1.2 Results

Of the sixty-two participants recruited, 7 were excluded from analyses due to a high false positive rate (6 greater than 25%) or slow response times (1 greater than 900 ms). Eight targets, mis-categorized as nonwords by more than half of participants, were removed (1 from transparent set, 7 from opaque; see Appendix A). Trials where no response was saved due to the response time limit being exceeded (49 trials) and trials with incorrect

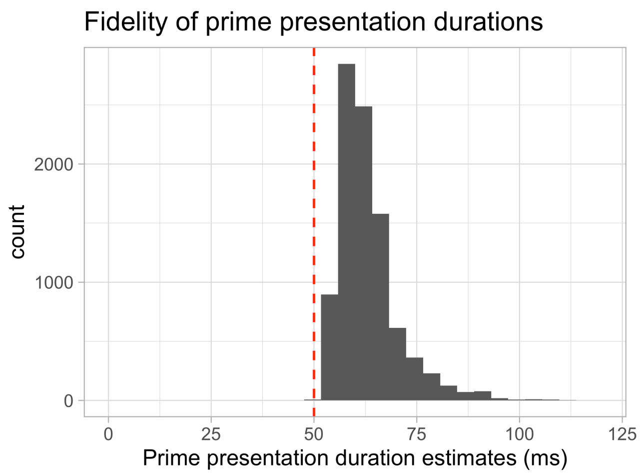

Figure 6.1: Histogram of prime presentation estimates for Study 1. The red dashed line denotes the intended prime duration (50 ms).

responses (1,050) were removed, as were reaction times less than 200 ms (20) and outliers (more than 3.5 standard deviations from by-participant mean; 61).

Estimates of prime presentation durations were generated using the by-trial `jsPsych` variable for amount of time elapsed since the participant began the study (see Figure 6.1). Looking at how well actual prime presentation durations aligned with designated duration (50 ms), three additional participants were excluded for inconsistent prime presentation durations (mean estimates greater than 75 ms or less than 50 ms). The participant who reported being able to consistently identify the prime was also removed, as described above. One hundred twelve trials were removed for having an estimated prime presentation of greater than 80 or less than 20 ms.[2] This left 8,581 responses to 99 transparent and 93 opaque targets, from 51 participants, for subsequent analyses.

The stimulus set was designed such that OSC was maximally varying, and minimally correlated with potentially confounding factors, *within* the two transparency conditions; however, the distribution of OSC values differed significantly between the transparent and opaque conditions. Accordingly, model fitting and statistical tests were conducted separately for these two sets of prime-target pairs. All models were fit using the `lme4`

[2]See Chapter 7 for a more detailed analysis of prime presentation fidelity.

(Bates et al., 2015) and `lmerTest` (Kuznetsova et al., 2017) packages in R (Version 4.0.3; R Core Team, 2020). Following the recommendation of Lo & Andrews (2015), reaction times were not transformed but instead were fitted with generalized linear mixed effects models, using an Inverse Gaussian distribution and an identity link function to capture the relationship between the predictors and the dependent variable.

6.1.2.1 OSC and semantically transparent morphological priming

The primary question of interest in this study is how the OSC of a morpheme impacts morphological priming magnitudes, and thus the interaction of OSC and priming magnitude is focal. However, as discussed in Chapter 5, target frequency appears to moderate the relationship between the newly calculated cluster-centered OSC and morphological priming magnitudes. Thus, the first model assessed for target responses in the semantically transparent condition included the three-way interaction of OSC, priming (related or unrelated) and target frequency. Morphological family size and target length were included as covariates (family size and target frequency were both taken from CELEX; Baayen et al., 1995), and the maximal random effects structure was used (Barr et al., 2013), including intercepts and slopes relative to the priming predictor for both participants and targets.[3] The three-way interaction was not found to be significant ($\beta = 20.12$, $p = 0.11$), so the model was re-fit with a two-way interaction of OSC and priming, including target frequency, morphological family size and length as covariates. The maximal random effect structure in this case only included random intercepts by participant. The simplified model showed a significant interaction of OSC and priming ($\beta = 34.05$, 0.017) as well as a significant main effect of priming ($\beta = 31.53$, $p < 0.0001$), but no main effect of OSC ($p > 0.5$). See Figure 6.2 for a visualization of this relationship.

[3]Most models did not converge using the maximal random effects structure: in these cases, random effects were simplified until the model did converge, following recommendations of Brauer & Curtin (2018).

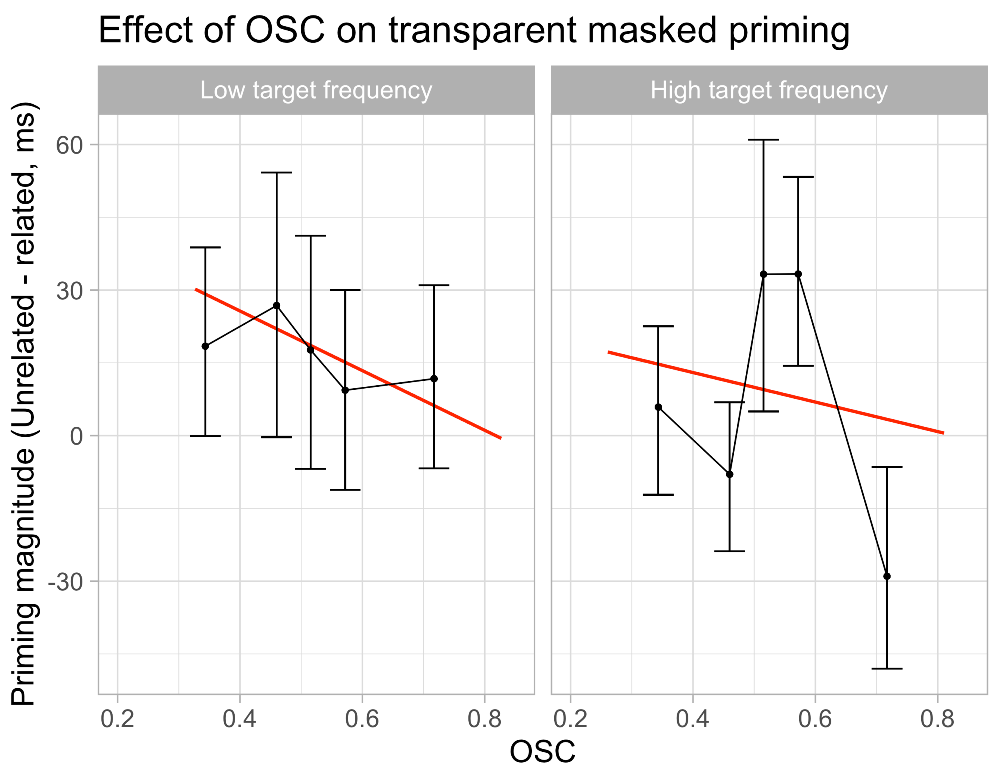

Figure 6.2: Relationship between cluster-centered OSC and transparent masked priming magnitudes in Study 1, median-split by target frequency. Each point represents mean priming magnitude by OSC quintile; error bars represent bootstrapped 95% confidence intervals. Red line shows line fit for by-item priming magnitudes. There was a significant effect of OSC on priming magnitude, and no significant interaction with target frequency.

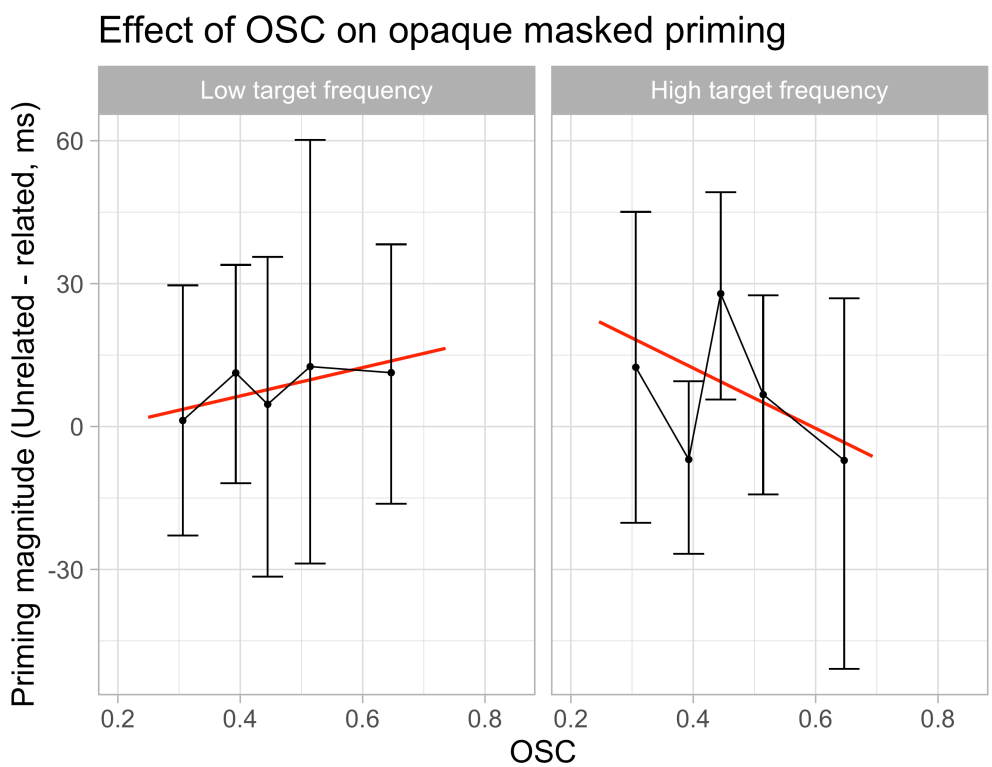

Figure 6.3: Relationship between OSC and opaque masked priming magnitudes, moderated by target frequency, in Study 1. The interaction with target frequency was significant in this case. See Figure 6.2 caption for details of figure generation.

6.1.2.2 *OSC and semantically opaque morphological priming*

Once again, the first model fit to responses for the semantically opaque prime-target pairs included a three-way interaction of OSC, priming, and target frequency, with family size and target length as covariates, and random slopes and intercepts for targets and participants. In this case, the three-way interaction was significant ($\beta = 44.47$, $p = 0.013$), as was the interaction of OSC and priming ($\beta = -86.56$,$p = 0.0028$), the main effect of priming ($\beta = 36.42$, $p = 0.015$), and the main effect of OSC ($\beta = 155.82$, $p < 0.0001$). This pattern of effects is depicted in Figure 6.3.

6.1.2.3 *Post-hoc comparisons for semantically opaque pairs*

To further investigate the nature of the OSC-priming interaction for lower-frequency targets (as shown on the left in Figure 6.3), models were fit to the lower-frequency and higher-frequency targets separately (median split). Target frequency was removed as a covariate, and the random effects maximal structure in both cases was random intercepts by participant. The interaction of OSC and priming magnitude was marginally significant for higher-frequency targets ($\beta = 38.56$, $p = 0.10$) and not significant for lower-frequency targets ($\beta = 21.44$, $p = 0.59$). It is worth noting that in these models, a positive interaction of OSC and priming signifies a reduction in priming magnitude with larger OSC values (as reaction times for related prime-target pairs are considered relative to unrelated pairs). Thus the direction of the marginally significant interaction for high-frequency targets aligns with the relationship depicted in Figure 6.3, while that of the non-significant interaction for low- frequency targets does not.

6.1.3 Discussion

The cluster-centered OSC metric is intended to quantify the degree to which a given letter string is semantically useful; in other words, how far on the spectrum from morpheme to non-morpheme does it fall? Given this view, we would expect morphological priming magnitudes to increase for prime-target pairs sharing higher OSC letter strings. This is not what we observe in the present study: OSC shows an overall negative effect on transparent priming magnitudes, and in the case of opaque priming, OSC interacts with target frequency to show neutral and negative effects for low- and high-frequency targets, respectively. How can we reconcile this pattern of effects with our conceptualization of morphological processing?

One perspective to consider is that this metric is orthographically-informed, and participants are all skilled adult readers. With experience, the process of recognizing a whole word form and retrieving its meaning becomes more and more efficient, and semantic and morphological information about the prime is accessed earlier. Particularly given that the prime presentations were overall slightly longer than intended (mostly falling between 55 and 70 ms), it is possible that the transparent prime is sufficiently processed for its semantic relation to the prime to have come into effect. If this is the case, a salient high-OSC substring within the prime may be a source of processing conflict, detracting from the prime's processing of whole-word characteristics that would most facilitate target recognition. A comparable interaction of stem- and word-level characteristics was demonstrated by Baayen et al. (2007) with prime and stem frequency: for high-frequency primes, stem frequency is inversely related to reaction times. The fact that there is a trend toward an interaction with target frequency ($\beta = 20.12$, $p = 0.11$) suggests the inhibitory effect of OSC is a bit weaker for higher-frequency primes. This makes sense as the amount of priming that can occur is constrained for high-frequency targets, as responses for these words are already quite fast.

Rapid access to semantic and morphological information about the prime would only clarify the pattern of results seen for transparent priming, not opaque priming. The negative effect of OSC on opaque priming magnitudes for high-frequency targets may have more to do with ceiling effects: given that high-frequency targets are already recognized quickly, ones that also have high OSC may be responded to even more quickly and thus be more difficult to prime. In the case of low-frequency opaque primes, a positive effect of OSC on priming magnitude could be canceling out or overcoming the increasing difficulty of priming higher-OSC targets, resulting in a neutral overall effect.

Despite these potential explanations, it is evident that the relationship of OSC and priming magnitudes is not a simple one, and the current pattern of results do not suggest that OSC generally boosts morphological priming. To tease apart what aspects of these results may be unique to the masked priming paradigm (and, correspondingly, to lower-level and more orthographically informed phases of word processing), we repeated this study with an overt, unmasked prime.

6.2 STUDY 2: THE EFFECT OF OSC ON OVERT MORPHOLOGICAL PRIMING

In this follow-up study, we look at how OSC interacts with both opaque and transparent morphological priming in the context of an *overt* prime, shown for 230 ms with no preceding mask.

6.2.1 Methods

6.2.1.1 Participants

Fewer participants were needed for this study, as patterns of overt priming are notably easier to detect than those of masked priming. Forty-three undergraduates were recruited from the same private university in Western Pennsylvania. As in Study 1, all participants were required to be native speakers of English with normal or corrected-to-normal vision, and they received course credit for participation.

6.2.1.2 Stimuli

Stimuli were identical to those used in Study 1.

6.2.1.3 Procedure

The procedure for this study was identical to that of Study 1, with the following exceptions:

- No hash mask was presented on any trial; instead, the prime word immediately followed the fixation cross.
- The prime was shown for 230 ms instead of 50 ms.
- Instructions emphasized that participants should categorize the second, lowercase string as a word or nonword, as the prime was now overtly visible.

- Participants were not asked about whether they noticed the prime word after finishing, as primes were presented for significantly longer with the intention of being perceived.

As in Study 1, the whole study took 35 minutes on average, with roughly 9 minute blocks of lexical decisions between breaks.

6.2.2 Results

Of the 42 participants recruited, 11 with false positive rates over 25% and 1 with mean reaction time over 900 ms were removed from further analysis. Twelve targets were removed for being responded to incorrectly by more than 50% of participants: 1 from the transparent condition and 11 from the opaque condition (see Appendix A). Trials where no responses was saved due to the response time limit being exceeded (70 trials) and trials with incorrect responses (600) were removed, as were reaction times less than 200 ms (1) and outliers (more than 3.5 sds from by-participant mean; 20).

Looking at prime presentation duration estimates, no additional participants needed to be excluded due to inconsistent prime presentation durations (all mean estimates between 200 and 300 ms). Sixty-one trials were removed for having an estimated prime presentation of greater than 300 ms or less than 200 ms. This left 4,889 responses to 99 transparent and 89 opaque targets, from 30 participants, for subsequent analysis.

Statistical models were fit following the same approach described for Study 1: separately for opaque and transparent pairs, using generalized linear mixed effects models with Inverse Gaussian distribution and identity link function.

6.2.2.1 OSC and semantically transparent morphological priming

The model of transparent morphological priming reaction times included a three-way interaction of OSC, priming, and target frequency, with morphological family size and target length as covariates, as in Study 1. The maximal random effects structure included

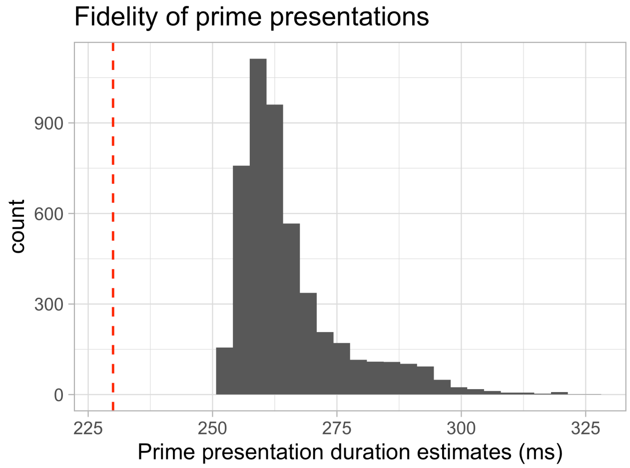

Figure 6.4: Histogram of prime presentation estimates for Study 2. The red dashed line denotes the intended prime duration (230 ms).

random intercepts by target and random intercepts and slopes relative to priming by participant.

The three-way interaction of OSC, priming and target frequency was significant ($\beta = -70.18$, $p < 0.0001$) as was the overall effect of OSC on priming magnitudes ($\beta = 264.50$, $p < 0.0001$) and the main effects of OSC ($\beta = -73.45$, $p = 0.022$) and priming ($\beta = -210.64$, $p < 0.0001$). See Figure 6.5 for a visualization of this pattern of effects.

6.2.2.2 *OSC and semantically opaque morphological priming*

Responses to opaque prime-target pairs were also fit with a model including the three-way interaction of OSC, priming, and target frequency, with family size and target length as covariates. The maximal random effects structure matched that for the transparent priming model, including random intercepts by target and random intercepts and slopes relative to priming by participant.

The three-way interaction was not significant ($p > 0.35$); a new model fit with target frequency as a covariate instead of a moderator revealed no interaction of OSC and priming, either ($p > 0.5$). However, a model fit to examine the main effect of priming, with OSC, target frequency, family size and target length as moderators, revealed a significant

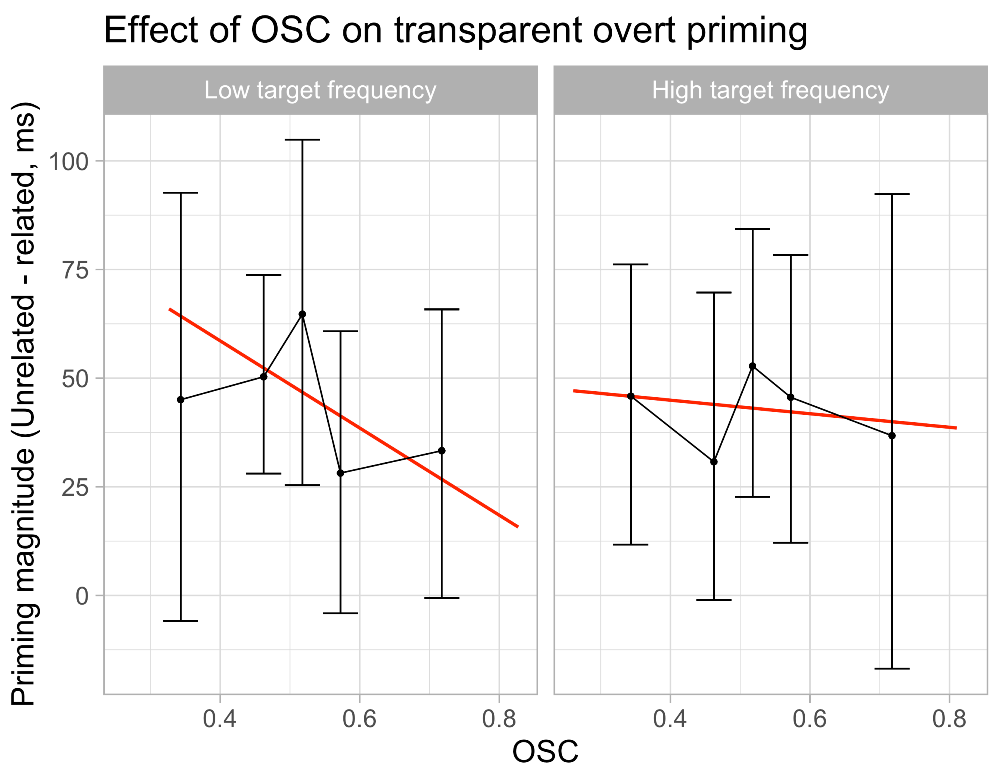

Figure 6.5: Relationship between OSC and transparent overt priming magnitudes, moderated by target frequency, in Study 2. The three-way interaction was significant. See Figure 6.2 caption for details of figure generation.

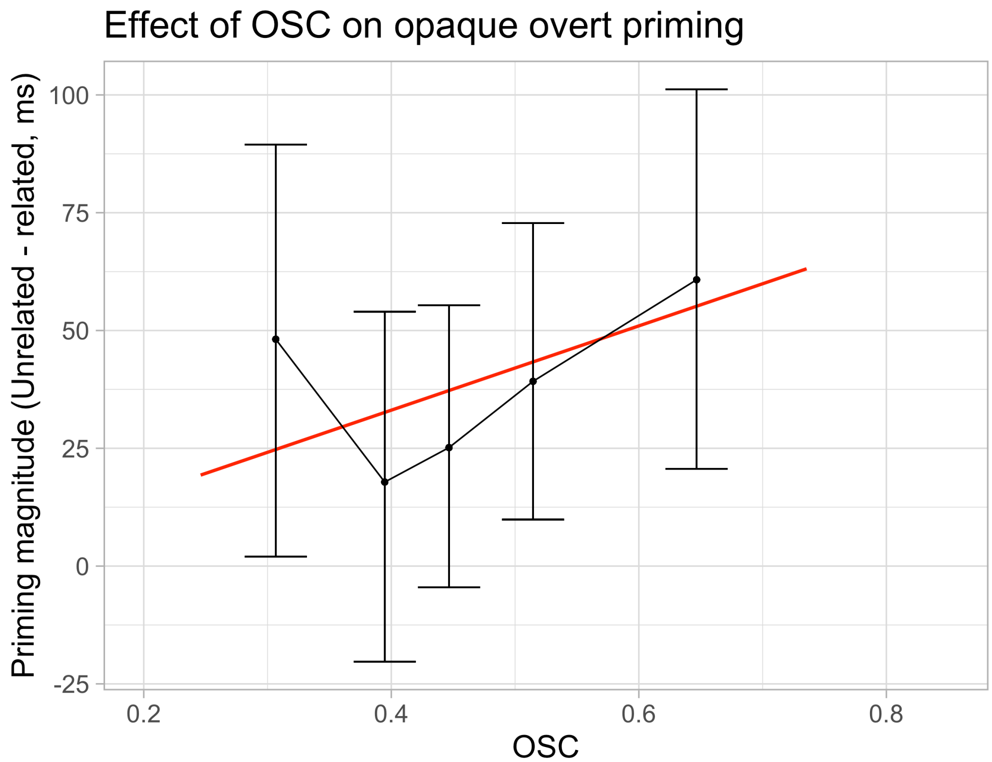

Figure 6.6: Relationship between OSC and opaque overt priming magnitudes in Study 2. The interaction of OSC and opaque priming was not significant, whether moderated by or unmoderated by target frequency. See Figure 6.2 caption for details of figure generation.

effect of opaque morphological priming ($\beta = -37.83$, $p < 0.0001$). See Figure 6.6 for a visualization of this pattern of effects.

6.2.3 Discussion

Once again, we see a negative relationship between transparent priming and OSC. It's possible that this effect is still due to an interference effect of stem OSC, detracting from beneficial whole-word processing, as discussed in relation to Study 1. However, the pattern in this case could also be explained entirely by target ceiling effects. Given that transparent primes are shown for longer in Study 2, priming may already be quite large regardless of OSC. Thus, for low-frequency targets, increasing OSC pushes target response times towards the response speed limit, due to this metric's significance for the target as opposed to the prime-target relationship, reducing room for priming to occur. High-frequency targets following overt transparent primes are already closer to that limit, and thus the negative effect of OSC on magnitudes is weaker for these targets, as there is less reduction in priming magnitude to be had.

Given the nature of the OSC metric, capturing a property of the stem across all its occurrences with no regard for morphological relatedness or semantic context, it seems unlikely that stem OSC should interfere with prime processing once the prime is presented for a full 260 milliseconds (given that prime presentations were slightly longer than intended). Additionally, this explanation is supported by the statistical model for the transparent overt data: target frequency had a strong negative main effect on reaction time (-35.79 ,$p < 0.0001$), whereas this effect was notably weaker for opaque overt priming data (For the three-way interaction model: 13.59, $p > 0.3$; for the main effect model: -14.59, $p = 0.04$).

OSC did not have any reliable effect on priming magnitudes in the case of opaque overt priming, regardless of target frequency. If our ceiling effects explanation for the transparent pattern of overt priming effects is correct, lack of an OSC effect here is probably due to the fact than an opaque prime is not as facilitatory as a target prime, and thus the additional constraint of target OSC on priming magnitudes is not detectable in this case. (This assumes a non-linear relationship between ceiling effects due to target OSC and those due to target frequency.) As noted above, OSC was not expected to facilitate priming in this case because it is a more orthographically-informed metric whose information would presumably be overrided by the prime's whole-word information after 230 ms (250 ms) of display time.

We did, however, observe a significant main effect of opaque priming, suggesting that sensitivity to opaque primes can in fact be detected in an overt priming context. This is surprising because opaque overt priming is typically found only in morphologically rich languages like Hebrew (Frost et al., 2000), Arabic (Boudelaa & Marslen-Wilson, 2001), and German (Smolka et al., 2015), and not in English (Rastle et al., 2000). Indeed, that expectation formed the premise of the modeling work presented in Chapter 4.

6.3 DISCUSSION

The two empirical studies reported in the current chapter examined the effects of OSC on morphological priming in adults, both under brief masked conditions (Study 1) and under overt, unmasked conditions (Study 2). Although results were somewhat difficult to interpret, this sequence of studies afforded several insights which will inform subsequent chapters of this thesis. First, patterns of results undeniably demonstrate sensitivity to the relatedness of the prime, even when it is masked, supporting this remote paradigm as a viable means of investigating masked priming effects. However, prime presentation times appear to be noisy, and participants are less motivated to stay on task (inferred from their slower reaction times and lower accuracy), so data cleaning and analyses must be conducted very carefully.

Second, the only OSC effects on priming magnitude that were observed were negative. We suspect that this finding has more to do with the effect of OSC on recognition of the target, and resulting ceiling effects, than with the effect of OSC on the prime-target relationship. The fact that the pattern of effects for transparent prime-target pairs is the same in both masked and overt priming contexts gives us reason to believe that the masked transparent effects can actually also be attributed to ceiling effects, instead of the stem versus whole-word interference explanation provided in the discussion for Study 1. Thus, facilitatory effects of OSC on morphological priming, if they exist, may be quite difficult to detect.

Taking an optimistic perspective, we note that the context in which we would most expect to see a facilitatory effect of OSC, opaque masked priming of low-frequency targets, did not show a negative effect. It is possible that in this case, a positive effect of OSC on morphological priming interacts with the ceiling effects described, resulting in no detectable effect. The issue of ceiling effects is certainly a challenge for interpreting the role of OSC; one possible solution is to investigate their effects in a population whose reaction times for lexical decision are slow. This, combined with the predicted role of OSC in acquiring morphological sensitivity, motivates the exploration of OSC effects on morphological priming in developing readers in Chapter 8.

Third, we found strong evidence of overt, opaque priming in Study 2. As mentioned, this is a surprising effect to encounter in a less morphologically rich language like English, and contradicts the results of Rastle et al. (2000). Given that there was no effect of OSC, the fact that stimuli were selected to maximize variance in OSC is unlikely to explain why opaque overt priming was found in this study and not in others. It's worth noting that only a few studies have demonstrated a lack of overt opaque priming effects in English. In addition to Rastle et al. (2000), Feldman et al. (2004) found an effect of semantic transparency on priming magnitudes and no effect of opaque priming for an overt visual prime (200 ms), and Feldman & Soltano (1999) found opaque and transparent priming for a 48 ms prime but only transparent priming for a 250 ms prime. Both of these studies used complex prime-target pairs with a shared stem or pseudo-stem (e.g., ACCORDANCE-ACCORDING versus ACCORDANCE-ACCORDION), as opposed to a complex prime paired with simple target (ACCORDANCE-ACCORD) as is done in the majority of morphological priming studies. Rueckl & Aicher (2008) also investigated semantic transparency effects, but in the context of long-term priming, which would be expected to differ from immediate priming. Besides these, most studies investigating semantic transparency effects on overt priming have been cross-modal, with the prime presented aurally (e.g., Marslen-Wilson et al., 1994; Gonnerman et al., 2007), and the patterns of effects may well differ when primes are visually or aurally presented. The main effect of overt priming in Study 2 suggests that it may be premature to conclude that there is no overt visual priming for opaque prime-target pairs. Given the central nature of opaque priming to the controversy between distributed and decomposition viewpoints, and consequently its central nature to the current thesis, we decided to investigate the robustness of the results reported by Rastle et al. (2000) with a close, remote replication study, reported in the next chapter.

CHAPTER 7

Clarifying the Time-Course of Morphological Processing in Skilled Adult Readers

Understanding how represented information changes over the time-course of visual word recognition provides a window into the mechanisms of this process. As discussed in Chapters 2 and 4, varying the duration for which a prime is presented prior to a target is one approach to identifying at what point in processing different prime-target relationships become more relevant, or in other words what aspect of the information carried by the prime is currently most dominant in how it is being represented.

The findings reported in Rastle et al. (2000) provided a beneficial overview of patterns of effects due to various word characteristics as a word is recognized, by assessing priming magnitudes for three different prime presentation durations: 43 ms, 72 ms, and 230 ms. They found strong and consistent priming by identity (STRONG-STRONG) and morphologically transparent (TEACHER-TEACH) pairs across all three SOAs, while orthographic priming (CASHEW-CASH) diminished and semantic priming (SKIP-MISS) strengthened with longer prime durations (see the left side of Figure 7.2 for a visualization of effects reported in this study). These patterns align with our intuition that earlier word processing is more sensitive to orthographic factors such as shared letters, while later word processing is more attuned to semantic characteristics. The pattern of effects for semantically opaque morphological prime-target pairs was particularly interesting, in that opaque priming was similar in magnitude to transparent priming for 43 ms SOA, but decreased to be comparable to orthographic priming at subsequent SOAs. Though opaque priming was not found to be significantly stronger than orthographic priming, it did diminish significantly with respect to transparent priming as SOA increased while the same interaction was not significant for orthographic with respect to transparent. This evidence for a short-lived sensitivity to semantically opaque morphological primes, more directly presented by Rastle et al. (2004), sparked new speculation regarding early stages of complex word processing.

The current chapter describes a close replication of Rastle et al. (2000), to assess the robustness of this pattern of results. Given the implications of these results for larger-scale interpretations of how word processing dynamics unfold, it is important to know whether they replicate, particularly given our reported findings of positive overt priming by semantically opaque morphological primes in Chapter 6. The most notable difference between this study and Rastle et al. (2000) is the fact that the current study was conducted remotely: thus, the current chapter also provides an opportunity to assess how similar results for remote masked primed lexical decision studies are to in-person instances with otherwise very similar stimuli and procedures.

7.1 METHODS

7.1.1 Participants

Participants were 96 undergraduate students at a private university in Western Pennsylvania, who received course credit in exchange for completing the study. All participants had normal or corrected-to-normal vision, and were native speakers of English. More participants were recruited than in the original study (24 more participants overall, 8 more per SOA), primarily due to the unmoderated remote nature of data collection as well as noisier stimulus presentation (demonstrated in Chapter 6).

7.1.2 Stimuli

Experimental prime-target pairs were taken directly from Rastle et al. (2000). This included 24 pairs in each of five conditions: 1) semantically transparent morphological, referred to as "morphologically, semantically, and orthographically related", or [+M+S+O], in Rastle et al. (2000); 2) semantically opaque morphological [+M-S+O]; 3) semantic [-M+S-O]; 4) orthographic [-M-S+O]; and 5) identical (e.g., cape-CAPE). During original stimulus selection by Rastle et al. (2000), morphological relationships between prime and target were established using the Oxford English Dictionary, and semantic relationships were confirmed using participant ratings and LSA ratings (Landauer & Dumais, 1997).

The exact unrelated primes, filler words, and nonwords used by Rastle et al. (2000) were not provided; as such, these items were generated following their described process. For each of 120 targets, an unrelated prime was selected from the SUBTLEX-US vocabulary (Brysbaert & New, 2009) that had no semantic or morphological relation to the target, matched the related prime in length, and did not share letters in the same position as the target. Sixty filler primes and targets with word lengths within the range of experimental primes and targets, frequencies were sampled from SUBTLEX-US, to reach a 30% ratio of related to unrelated primes among word targets.

One hundred eighty pairs with nonword targets were generated. Twenty four nonword targets were pseudo-stems of real words (e.g., BANT from BANTER or LEOP from LEOP), paired with their source word. One hundred fifty-six pronounceable nonwords were selected from the English Lexicon Project, length-matched to the real word targets (Balota et al., 2007); 24 were primed by themselves (slint-SLINT) and the rest were paired with an orthographically unrelated word, length-matched to real word primes (e.g., slither-BRAFT). This resulted in 360 trials for lexical decision: 180 nonwords and 180 words.

7.1.3 Procedure

Thirty-two participants were assigned to each of the three SOA conditions. In each SOA, 16 participants were presented with list 1 and 16 were presented with list 2.

As in the studies presented in Chapter 6, data collection was remote and unmoderated. Stimuli were presented using `jsPsych` (De Leeuw, 2015), with the `jspsych-psychophysics` plugin (Kuroki, 2021) for masked priming trials presentation and the `jspsych-virtual-chinrest` plugin to accommodate participants' varying screen sizes (Li et al., 2020).

Participants were instructed to decide for each target string whether it was a word

("Press the J key") or not ("Press the F key"). Participants assigned to the 230 ms SOA were told that they would briefly see a lowercase string before the uppercase string, and to only respond to the uppercase string. Participants assigned to the 43 ms and 72 ms SOAs were not informed of the presence of the lowercase prime. Sixteen practice trials were administered prior to beginning the main experiment. Participants were given feedback on the speed and accuracy of their practice performance, and given the option to repeat instructions and practice trials. The mean reaction time threshold for recommending faster responses was 720 ms, and participants were advised to review the instructions if their accuracy was below 85%.

Lexical decision trials began with a fixation cross in the center of the screen for 500 ms, followed by a mask of hash marks (########) for 500 ms. The prime word was then flashed for the specified time (set to 43, 72, or 230 ms) and immediately followed by the target word. The target word was shown until a response was given, or until the trial ended if no response was provided after 2 seconds of target display. Participants were informed of the time limit for responses during instructions. A blank screen was shown for one second between trials.

The study took 23 minutes to complete, on average. Lexical decisions were split into three 6-minute chunks with optional, 30 second breaks in between.

As in the masked priming study in Chapter 6, participants in the 43 and 72 ms conditions were asked about their experience of the flashed prime word. Sixty-three percent of participants in the 43 ms condition reported that they didn't notice anything else besides the fixation cross, hash mask, and target word. Of the 12 who did report noticing something, 9 noticed that the flashed stimuli was a character string of some sort, but all said they either never recognized the word or only recognized the word on a few trials. For the 72 ms condition, 41% of participants didn't notice the prime stimulus. Of the 19 who did, all noticed that the flashed stimulus was a word, 10 reported seeing it in all or most trials, and 4 reported being able to recognize the flashed word most of the time. As the 72 ms SOA is on the edge of prime durations we would expect to be fully masked, the participants who reported being able to recognize the word were not excluded from main analyses.

7.2 RESULTS

The analyses described below address both objectives outlined in the introduction to this chapter: to examine the robustness of the time-course effects reported by Rastle et al. (2000), and to assess the soundness of remote data collection for investigating morphological processing effects with the primed lexical decision paradigm.

7.2.1 Data cleaning

The approach to trimming data was slightly different from that of Rastle et al. (2000), as participant responses were overall slower and less accurate in the current study. This difference can likely be attributed to the remote context of data collection. One participant was removed for having a mean reaction time (RT) of over 1000 ms (this threshold was 800 ms in the original study), and 8 participants were removed for having a false positive rate of over 25% (original cut-off was 20%). This left 87 participants: 33, 26, and 28 for the 43 ms, 72 ms, and 230 ms SOAs, respectively. Reaction times were removed if the response was incorrect or not recorded due to exceeding the trial time limit (632 responses), or if they were greater than 1550 ms (20 responses, matching the removal of 0.25% of responses using a cut-off of 1400 ms in Rastle et al. (2000)). The recorded data was then examined to determine how faithfully prime presentation times on participants' computer screens adhered to their assigned duration (43, 72 or 230 ms).

7.2.2 Checking fidelity of prime presentation durations

The `JsPsych` package saves the average number of milliseconds spent on each frame (`avg_frame_time`) and the number milliseconds since the participant began the study

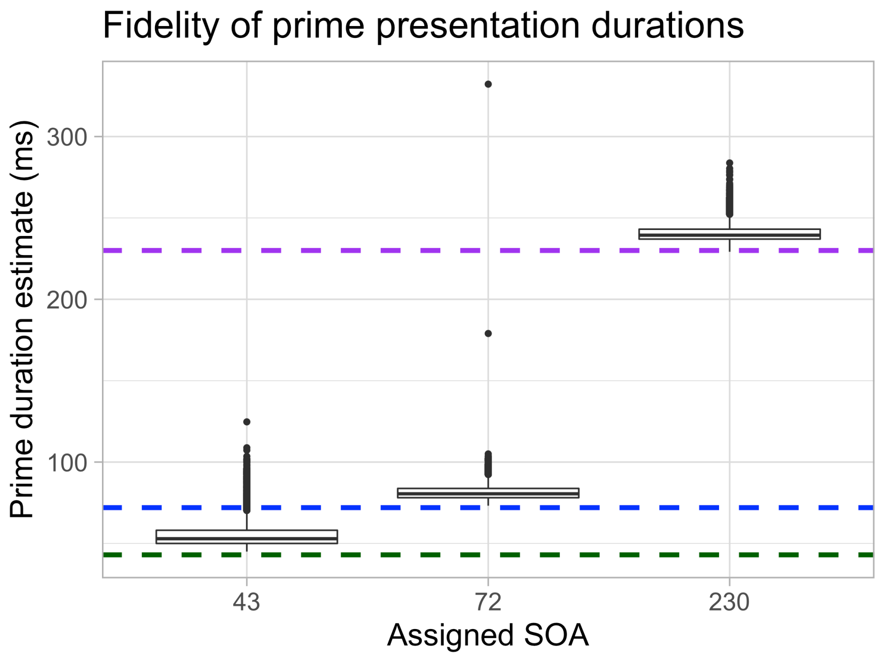

Figure 7.1: Confirmation that prime duration estimates mostly align with the desired value across SOA conditions. Trials with prime duration estimates more than 25 ms from the desired value (e.g., prime estimates greater than 68 ms in the 43 ms SOA) were removed from analysis.

(`time_elapsed`) for each trial (De Leeuw, 2015). The first variable was used to calculate the average duration of a single display frame for each computer used to complete the study. All participants' frame durations were at or below 16.67 ms except one individual in the 43 ms SOA (mean frame duration of 33.3 ms), who was removed from further analyses (leaving 86 participants total). Such a slow refresh rate would make it impossible for this individual's computer to reliably differentiate between 43 and 72 ms presentation durations.

Prime presentation duration was then estimated for each trial by using `time_elapsed` to infer trial duration, and then subtracting the amount of time needed to display pre-prime stimuli (2000 ms) and recorded reaction times. The resulting estimates of prime presentation times for each SOA are shown in Figure 7.1. Trials with prime duration estimates more than 25 ms from the assigned value (e.g., prime estimates greater than 68 ms for the 43 ms SOA) were removed from analysis, resulting in the removal of 11.5%, 4.5%, and 4.9% of trials from the 43 ms, 72 ms, and 230 ms SOAs, respectively.

Prime durations in the 43 ms SOA condition, though mostly concentrated around 50 ms, trended towards being shorter than those in the masked priming study with prime durations set to 50 ms, presented in Chapter 6 ($\beta = -3.95$, $p = 0.081$). For the 43 ms SOA, 2,696 prime estimates were less than 50 ms, relative to only 3 prime estimates less than 50 ms for the 50 ms study. Given that these values are estimates (relying on the assumption that all other stimuli — e.g., fixation cross, hash mask — are presented for exactly the intended duration), there may have been a higher proportion of trials for which primes were presented too briefly to be processed in the current study than in the prior study (this possibility is returned to in the discussion).

Applying the filters for speed, accuracy, and display precision described above resulted in 8,767 reaction times for use in subsequent analyses.

7.2.3 Main analyses

Following the analyses described in Rastle et al. (2000), the data was analyzed using a mixed-design ANOVA with four factors: SOA (3 levels), condition (5 levels), list (2 levels), and priming (2 levels). For the by-participants model, condition and priming were repeated factors; for the by-items model, SOA and priming were repeated factors.

The main effect of priming was significant for both the by-participants model ($F_p(1, 85) = 97.45$, $p < 0.0001$) and the by-items model ($F_i(1, 115) = 45.47$, $p < 0.0001$), indicating that response times for words presented after related primes differed significantly from those presented after unrelated primes. The interaction of condition and priming was also significant for both models ($F_p(4, 336) = 6.42$, $p < 0.0001$; $F_i(4, 111) = 3.21$, $p = 0.016$),[1] as was the three-way interaction of condition, priming, and SOA ($F_p(8, 328) = 2.90$,

[1]For all by-participants models other than those isolating main effects of priming, one participant was removed from analysis because they did not have trials for all combinations of condition and priming. Thus, results from these models based on 85 as opposed to 86 participants.

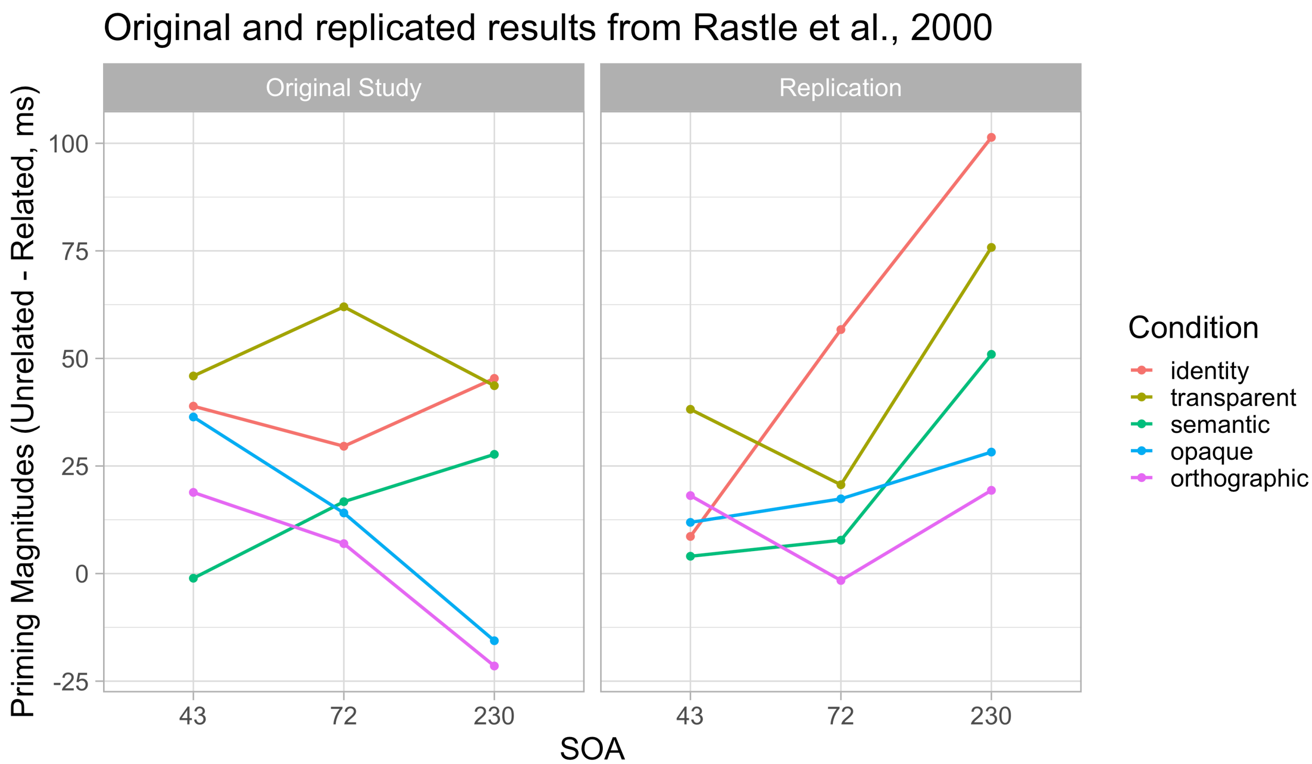

Figure 7.2: Comparison of priming magnitudes across condition and SOA from Rastle et al. (2000)] and the current, close replication study. The left axis represents amount of priming, calculated as mean reaction time for a target following a related prime, subtracted from mean reaction time for that target following an unrelated prime.

$p = 0.0039$; $F_i(8, 222) = 2.74$, $p = 0.0067$). There was no significant four-way interaction of condition, priming, SOA and list (both ps > 0.5). Thus, priming magnitudes varied significantly depending on the relationship between prime and target (transparent, orthographic, etc.) and this pattern of effects depended on the duration of prime presentation. See Figure 7.2 for patterns of effects found in the current study compared with those of Rastle et al. (2000).

A series of planned contrasts further illuminated the nature of the three-way interaction of priming, condition, and SOA:

- Looking only at the transparent and identity priming conditions, the interaction between priming and condition was not significant (both ps > 0.25), but there was a three-way interaction of priming, condition, and SOA ($F_p[2, 82] = 7.40$,$p = 0.0011$; $F_i[2, 88] = 5.16$, $p = 0.0076$), signifying that the difference between transparent and identity priming varied with SOA. This interaction differs from findings in Rastle et al. (2000), and likely captures the flip from stronger transparent priming to stronger identity priming from the 43 ms to the 72 ms prime duration. When data from participants who were shown primes for 43 ms were removed from this contrast, the three-way interaction was no longer present (both ps > 0.35) and the interaction of priming and condition became significant ($F_p(1, 53) = 9.87$,$p = 0.0028$; $F_i(1, 44) = 4.53$, $p = 0.039$), confirming that identity priming was consistently stronger than transparent priming for 72 ms and 230 ms SOAs.

- For transparent and semantic conditions, an interaction of condition and priming (significant by-participants: $F_p[1, 84] = 8.22$, $p = 0.0053$; marginal by-items: $F_i[1, 46] = 3.39$,$p = 0.072$) was found, indicating that transparent priming was overall stronger than semantic priming. The three-way interaction of priming, condition and SOA was not significant (both ps > 0.45). These results match findings from Rastle et al. (2000).

- Comparing transparent and orthographic conditions, there was a significant interaction of priming and condition ($F_p[1, 84] = 14.61$, $p = 0.00025$; $F_i[1, 45] = 5.93$, $p = 0.019$) such that transparent priming is overall stronger than orthographic priming. The three-way interaction of priming, condition, and SOA did not reach significance ($F_p[2, 82] = 2.02$, $p = 0.14$; $F_i[2, 90] = 1.93$, $p = 0.15$). Both results align with those of Rastle et al. (2000).

- Considering the transparent and opaque conditions, transparent priming was stronger than opaque priming overall (significant by-participants: $F_p[1, 84] = 6.43$, $p = 0.013$; marginal by-items: $F_i[1, 45] = 3.62$, $p = 0.064$), but this difference did not vary with SOA ($F_p[2, 82] = 1.11$, $p = 0.34$; $F_i[2, 90] = 1.82$, $p = 0.17$). Rastle et al. (2000)

similarly found stronger transparent priming than opaque priming overall, but also found opaque priming decreasing significantly relative to transparent priming with increasing SOA.

- For the opaque and orthographic conditions, a main effect of priming was found ($F_p[1,84] = 10.24$, $p = 0.0019$; $F_i[1,45] = 5.27$, $p = 0.026$) but neither the interaction of priming and condition nor that of priming, condition and SOA were significant (all ps > 0.25), mirroring results of Rastle et al. (2000). However, unlike in the original study, there was no change in priming magnitude with increased prime duration ($F_p(2,82) = 1.39$, $p = 0.26$; $F_i(2,90) = 0.88$, $p = 0.42$) for these conditions.

7.3 DISCUSSION

The current chapter reports an attempt to replicate Rastle et al.'s (2000) findings of the effects of SOA on morphological priming—and, in particular, opaque priming—but using remote testing. Overall, the primary differences between the results of Rastle et al. (2000) and those presented here are that, first, identity priming is oddly low for the 43 ms SOA and, second, opaque and orthographic priming do not decrease in magnitude with greater SOA. The identity priming difference may be due to the nature of remote masked priming presentation: overall, the effect of condition on priming magnitudes appears less dramatic than in the original results. With the masked prime set to 43 ms (instead of 50 ms as in Chapter 6), there may be a higher proportion of trials when the prime was shown too briefly to be processed, or no prime was actually shown, and our analysis procedures using the `time_elapsed` in `jsPsych` to estimate prime presentation times may not be precise enough to fully account for this. Alternatively, in the relaxed atmosphere of a remote study, participants may not be sufficiently focused to show consistent sensitivity to a 43 ms (as indicated by their slower responses and higher false positive rates relative to participants in Rastle et al., 2000). Either way, filtering for just the 43 ms trials in this study, the interaction of priming and condition showed only a trend toward significance ($F_p(4, 120) = 2.00$, $p = 0.099$; $F_i(4, 111) = 1.88$, $p = 0.12$). Given the larger sample size and more prime-target pairs per condition, this was not an issue for the masked priming study reported in Chapter 6, but it does suggest that the configuration of transparent and identity priming for 43 ms SOA should not be overinterpreted.

The lack of a decrease in opaque and orthographic priming is interesting, and was the primary purpose of the current replication study. These results suggest that, at least in the context of a complex word priming its simple stem (see Feldman & Soltano, 1999; Feldman et al., 2004, for alternatives), semantically opaque and orthographic effects in an overt visual priming context may be more facilitatory than the pattern found by (Rastle et al., 2000).

CHAPTER 8

The Role of Consistency in the Emergence of Morphological Processing in Developing Readers

The developmental literature on complex word reading primarily focuses on conscious reasoning about words' structure, referred to as *morphological awareness* (e.g., Nagy et al., 2006). This is a distinct area of research from morphological *processing*, which primarily focuses on processes occurring before the word has been recognized. Morphological processing research in developing readers is much more limited. In English, only two studies to our knowledge explore masked priming effects in young readers (Dawson et al., 2021; Beyersmann et al., 2012a), and these studies used the same set of stimuli. Both studies detected transparent morphological masked priming as early as 3rd grade (around 9 years old), whereas opaque morphological priming is not detected until late middle school or early high school (around 14 years old). Additionally, participants' word recognition skill better predicts when these effects emerge than does their age (Dawson et al., 2021).

As mentioned previously, several theories of morphological processing propose that morphological sensitivity emerges due to the detection of letter sequences that are associated with similar meanings across contexts, providing "islands of regularity" that can facilitate semantic access in some or most words in which they appear (e.g., Plaut & Gonnerman, 2000; Rastle & Davis, 2008). Thus, orthographic-semantic consistency (OSC) may be a useful metric to predict which morphemes impact visual processing earliest in developing readers.

In the current cross-sectional developmental study, readers from 4th to 9th grade (ages 9 to 16) completed a masked primed lexical decision task. Prime-target pairs were selected for maximally varying stem OSC, comparable to those used in Chapter 6, to allow examination of how the effect of OSC on priming changes with reading experience. Priming effects in less experienced readers are expected to exhibit greater sensitivity to the OSC of prime-target pairs' shared morpheme than those in more experienced readers, particularly in the case of opaque priming. In addition to providing insight regarding the emergence of morphological processing effects, the current study provides another opportunity to examine how (and whether) OSC interacts with morphological priming, in a sample that is potentially less susceptible to the ceiling effect issues encountered in Chapter 6.

The same stimuli were presented in an overt priming context to a separate cohort of 4th and 5th graders. This condition was included to more closely examine the time-course of morphological processing when sensitivity to morphology is less fully formed. French-reading children in 3rd grade have been found to show sensitivity to opaque primes (e.g., BAGUETTE, "breadstick", priming BAGUE, "ring") for both short (60 ms) and long (250 ms) prime duration (Quémart et al., 2011). However, 5th graders, and 7th graders, and adults show such priming only for the 60 ms prime duration. These findings contrast with results in English, where opaque masked priming is not detected until around 7th grade. The pattern of effects found in Quémart et al. (2011) raises the possibility that opaque priming does occur in English, but can only be detected for longer prime durations due to the slower processing of lexical information in less experienced readers. Additionally, the simulated developmental trajectory of opaque priming for the morphologically rich language in Chapter 4 supports the possibility that opaque priming emerges in overt priming context, or both masked and overt priming contexts, at first. Given that no other work on morphological processing acquisition has manipulated prime duration within the same study, we considered it worthwhile to include an overt condition for the youngest age group in the current design. An overt priming condition was not included for the older age groups (6th to 9th grade) due to greater difficulties recruiting enough participants in these age ranges.

8.1 METHODS

8.1.1 Participants

One hundred twenty-one participants between 4th and 9th grade were recruited for this study. This sample broke down as 59 4th and 5th graders (mean age: 10.2), 33 6th and 7th graders (mean age: 11.9), and 29 8th to 9th graders (mean age: 14.2). Participants had normal or corrected-to-normal vision, and all of them first learned to read in English. Four parents reported that their child had learned another language prior to English, but that their child had first learned to read in, and was currently more proficient at reading in, English. Given that recruitment was conducted via online advertisements as opposed to in a more uniform setting, such as a school, this sample was likely biased towards higher-performing readers for their age, who would be more willing to participate in a reading-related task in their free time.

Recruitment was conducted via Facebook (posting in groups related to parenthood, as well as sharing via personal pages) and the Children Helping Science website (https://childrenhelpingscience.com/). For 70% of participants, the parent met with the researcher via phone or video call to answer questions about their children's reading experience; the primary purpose of this call was to verify that they had a child who was eligible for the study.[1] Participants were sent a $10 gift card (via email to their parent) as a thank-you for their time and effort.

8.1.2 Stimuli

The process for selecting prime-target pairs was very similar to that described in Chapter 6: the Stochastic Optimization of Stimuli software (SOS!, Armstrong et al., 2012) was used to select items within each condition such that the OSC of the shared stem varied maximally, while remaining minimally correlated with prime and target frequencies, lengths,

[1] This protocol was put in place for all parents not personally known by the researcher to address issues with false and repeat participants, following difficulties in early data collection.

orthographic neighborhoods, and bigram frequencies, as well as stem and affix family sizes, and the number of words in which the stem was found embedded during calculation of OSC. The only characteristic significantly correlated with OSC in the final stimuli was target word length in the opaque set (0.38, $p = 0.015$). Correlations of orthographic target length and opaque prime length trended toward significance ($r = 0.21$, $p = 0.2$ and $r = 0.25$, $p = 0.12$, respectively), and no others were significant (all ps > 0.5).

The primary differences in the current stimuli selection procedure relative to that in Chapter 6 were: (1) the inclusion of an *orthographic* condition in addition to *transparent* and *opaque* morphological conditions, (2) the reduction of pairs per condition from 100 to 40 for a shorter experiment duration, and (3) the use of only higher-frequency targets (frequencies greater than 32 per million words according to the SUBTLEX-US frequency norms; Brysbaert & New, 2009), so that younger readers would be able to complete the task with sufficient accuracy. (See Appendix A for the experimental stimuli used.) Unrelated primes for each target were generated by shuffling related primes such that no unrelated prime had the same consonant onset as its target, or contained more than 40% of the letters present in the target. Each target was presented exactly once per participant, either with its related prime (e.g., BUZZARD-BUZZ) or with an unrelated prime (WALLOP-BUZZ); the pairing of each target with a related or or unrelated prime was counterbalanced across participants. Forty pairs of simple primes (5 to 10 letters long, to correspond with experimental primes) and simple targets (3 to 5 letters) were added as fillers, to reduce the ratio of real word targets presented with related primes to 37.5%. Filler primes and targets were identified as simple using the CELEX database (Baayen et al., 1995).

Sixty nonwords were created by swapping the onset consonant strings of experimental targets such that no nonword formed a word, including slang words (e.g., GOTH, RAD). These were primed by unused transparent, opaque, and orthographic primes from prior morphological priming studies, matched for length with experimental primes. Additionally, 20 nonwords were drawn from the English Lexicon Project (Balota et al., 2007) with lengths between 3 and 6 to be comparable with experimental targets, and paired with simple primes between 5 and 10 letters in length. Twenty pseudo-complex words and their nonword pseudo-stems (e.g., BANTER-BANT, KITCHEN-KITCH) were included as well.

Altogether, participants were presented with 260 trials: 160 words and 100 nonwords. This imbalance was tolerated for the sake of keeping overall experiment time as short as possible, while maximally varying OSC within each experimental condition. Trials were ordered such that no more than 5 trials went by without a nonword being presented.

8.1.3 Procedure

After returning the consent form and completing a 5-minute interview with the researcher, parents were provided with a participant ID and URL for their child, and instructed to ensure that the study be completed on computer or laptop, with at least 45 minutes available and minimal distractions. Parents were told that they may be in the same space when their child completed the task, but were asked not to "comment on how they are doing, give them answers, or watch their responses over their shoulder". As in Chapter 6, the entire experiment was presented using jsPsych (De Leeuw, 2015), with the jspsych-psychophysics plugin (Kuroki, 2021) for masked priming trials presentation and the jspsych-virtual-chinrest plugin to account for participants' screen size in determining stimulus font sizes (Li et al., 2020).

Study introduction and instructions contextualized the task as a means of defeating a super-villain intent on destroying the internet. Participants were told that, in order to discover a "master password," they would need to sort real words from fake words as quickly and accurately as possible.

A task designed to assess word recognition skill was administered first, followed by the main study, and finally the morphological awareness task. Participants were given four opportunities for optional, 30-second breaks: one after the word recognition measurement task, and one between four blocks of the primary experimental trials. During each break, their progress in cracking the code and defeating the evil villain was reported ("You've uncovered more of the password! So far you've uncovered: P L A _____"). The entire study took on average 33.5 minutes to complete.

8.1.3.1 *Assessing word recognition ability*

Prior to assessing developing readers' sensitivity to morphological priming, their word recognition skill was measured in order to provide a more accurate approximation of reading experience than age or grade level (see Dawson et al., 2021). One common approach to measuring word recognition ability is the Word Recognition in Isolation (WRI) task (Morris, 2013), in which the child reads aloud as many words as they can from a list of words spanning several reading levels within 90 seconds. Each word read and pronounced correctly adds a point towards their final word recognition score. To accommodate the remote and unmoderated data collection context, a variation on this task was devised that relies on lexical decision instead of reading aloud. Twenty-eight simple (i.e., non-complex) words spanning reading levels from grade 3 to grade 8 were selected from the WRI list, and 28 nonwords were selected from the English Lexicon Project (Balota et al., 2007) to match the words for length, orthographic neighborhood and bigram frequency. To make the overall experience of this set of lexical decision trials comparable to those of the morphological priming trials, a fixation cross (500 ms), a hash mask (800 ms), and a graphotactically illegal text string (XXOXOXX; 50 ms) were displayed before each target word appeared.[2] Targets were displayed for 5 seconds, and reaction times slower than 5 seconds were not recorded.

[2] Participants in the overt priming condition received an additional instructional interlude after this segment, informing them that they would now see two strings (one in uppercase before the one in lowercase to which they should respond) because the evil villain was trying to trick them by changing the task.

8.1.3.2 *Primary study: Masked priming*

Ten practice trials of the lexical decision task were administered prior to the main task. Each age group had its own speed threshold (1600 ms for 4th and 5th graders, 1000 ms for 6th and 7th graders, 750 ms for 8th and 9th graders), as more experienced readers would be expected to respond more quickly than less experienced readers. These thresholds were determined based on mean reaction times reported in Dawson et al. (2021). If responses were less than 80% accurate, or were on average slower than the threshold for that participant's age group, they were required to repeat the practice trials until they met criterion, or until they had repeated the practice three times[3].

Masked priming trials for the main experiment began with a fixation cross in the center of the screen for 500 ms, followed by a mask of hash marks (########) for 800 ms. The uppercase prime word was then flashed for 50 ms, and immediately followed by the lowercase target word. The target word was shown for 5 seconds or until a response was given, and reaction times slower than 5 seconds were not recorded. A blank screen was shown for one second between trials. This masked priming procedure closely follows the procedure used by Dawson et al. (2021), with the exception that primes were presented in uppercase rather than lowercase.

8.1.3.3 *Assessing morphological awareness*

At the end of the study, participants' morphological awareness was measured to ascertain whether children's ability to reason about words' morphological structure explains additional variance in morphological processing effects, after accounting for word recognition skill. Although it is possible that explicit knowledge of morphology may contribute to how children (and adults) process complex words, it was predicted that word recognition skill would be the stronger predictor of morphological priming magnitudes.

[3]Nine participants still had accuracy below 80% even after completing 3 iterations of practice, 7 of whom were in the youngest age group; this may indicate that the practice response speed threshold for the youngest age group was too fast.

Three subtests were selected from the University of Washington Language Battery, previously used with upper elementary and middle school students to investigate morphological awareness in this age range (Nagy et al., 2006). The first subtest was a sentence completion task with real words as response options; e.g., the options to complete the sentence “The doctor helps patients stay ___” were HEAL, HEALTH, HEALTHY, and HEALTHINESS (15 items). The second task was similar, but sentences were completed with pseudo-complex nonwords; e.g., the options to complete “The girl dances ___ ” were SPRIDDERISH, SPRIDDERED, SPRIDDERLY, SPRIDDING (4 items). For the third task, participants were asked to decide for pairs of words whether one of the words comes from the other (e.g., REASONABLE – REASON or DOLLAR – DOLL; 50 items). No word pairs presented in the third task matched prime-target pairs from the main experiment. These subtests were selected because they were most focused on word-level and not sentence-level reasoning of those available in the Washington Language Battery, and because they were easily adapted for remote and unmoderated data collection.

8.2 RESULTS

A good deal of data needed to be removed before primary analyses were conducted (unsurprisingly, given the remote and unmoderated nature of this developmental study). Twenty-seven participants of the original sample were removed due to having a false positive response rate of over 30%. Four prime-target pairs were removed because the target was mislabeled as a nonword by more than half of participants (3 from the orthographic condition, 1 from the opaque condition). Forty-eight trials were removed due to a lack of response within the 5-second response interval, and 969 were removed due to incorrect response. Fifty responses were removed as by-participant outliers (greater than 3.5 sds from mean reaction time), and 11 were removed due to being faster than 200 ms.

Prime presentation durations were estimated following the same procedure described in Chapters 6 and 7. Three additional participants were removed: one due to slow aver-

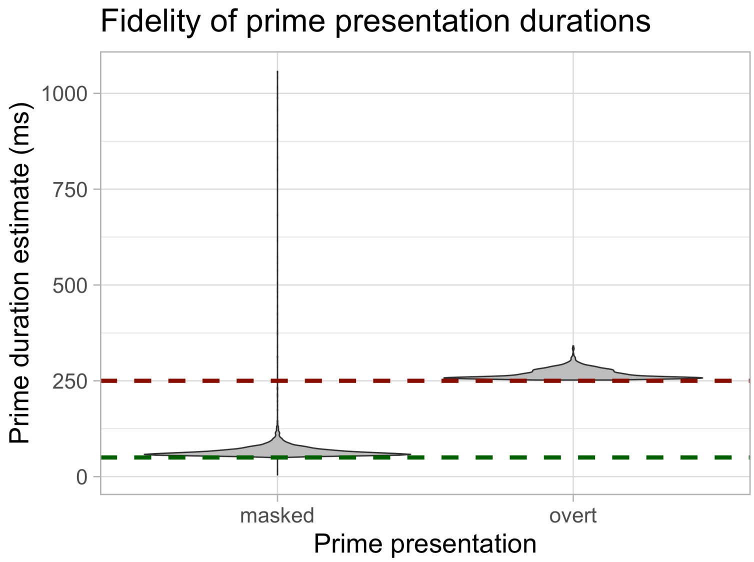

Figure 8.1: Estimates of prime duration for the two prime presentation conditions, prior to removal of excessively high or low estimates. Three times as many participants were assigned to masked priming than were assigned to overt priming.

age frame duration (over 30 ms) and two in the masked condition due to average prime durations greater than 100 ms. Trials with estimates greater than 90 ms or less than 10 ms were removed, for participants assigned to the masked prime presentation, and those with estimates greater than 300 ms or less than 200 ms were removed (see Figure 8.1). This resulted in a loss of an additional 3.2% of trials.

Following the data cleaning described above, 9,092 reaction times, in response to 116 prime-target pairs (37 orthographic, 39 opaque, and 40 transparent) from 90 participants remained for analysis. Twenty-two participants remained in the oldest group (8th and 9th grade), 28 in the middle group (6th and 7th grade), and 40 in the youngest group (4th and 5th grade), split between 21 viewing masked primes and 19 viewing overt primes.

8.2.1 Word recognition scores

Word recognition skill was estimated as accuracy of responses to real words. Accuracy was used instead of mean reaction time because it is a better estimate of participants'

Age Group	Prime Display	Lower WR	Higher WR
Younger	Masked	16	5
Younger	Overt	12	7
Middle	Masked	14	14
Older	Masked	4	18

Table 8.1: Assignment of participants to word recognition skill level from each age and prime display group, based on accuracy median-split.

word knowledge, particularly given the selection of vocabulary words spanning multiple grade levels, and because it is more distinct from the dependent variable (primed reaction times) that word recognition scores will be used to predict. Mean accuracy was 84.6%, and this metric was moderately correlated with participant age ($r = 0.40$, $p = 0.00010$).

Word recognition accuracy was used as the sole proxy variable for reading experience. This is due to the results of Dawson et al. (2021), in which their metric of word reading ability was found to significantly moderate the magnitude of opaque priming, while age did not. Although recruiting across a range of grade levels is an effective means of ensuring that word recognition skill varies sufficiently, a behavioral measure of word recognition serves as a more precise approximation of reading experience. Table 8.1 shows the distribution of participants in each age group that performed above or below the median word recognition score (79%). This table demonstrates that there were several high-performing readers in the youngest age group and similarly several low-performing readers in the oldest age group.

8.2.2 Changes in priming effects with reading acquisition

As in Chapter 6, all models reported in this and subsequent sections were fit to the transparent, opaque, and orthographic conditions separately because item characteristics were not matched across conditions during stimulus selection. Reaction times were fitted with generalized linear mixed effects models, with an Inverse Gaussian distribution and an identity link function. Due to concerns that certain effects were driven by outliers, model outliers (residuals more than 3.5 standard deviations from the mean) were removed and the model refit to the remaining data (Baayen & Milin, 2010). This data trimming never resulted in the removal of more than 0.7% of the data.

Morphological family size and target frequency were included as covariates in all models as they were found to be significant predictors of reaction times. Target length was also included in models of semantically opaque morphological priming which included OSC as a predictor, as target length and OSC were correlated for this subset of stimuli. All models included the maximal random effects structure (Barr et al., 2013).

For insight regarding how each type of priming changes with reading experience, the interaction of word recognition skill and priming (related versus unrelated) was assessed for each priming condition. For transparent morphological priming, there was no change in priming magnitude with word recognition skill ($\beta = 9.79$, $p > 0.6$). Removing the interaction term revealed main effects of both priming ($\beta = -23.55$, $p = 0.034$) and word recognition skill (-853.89, $p < 0.0001$). For opaque morphological priming, this interaction was significant ($\beta = -65.19$, $p = 0.0055$), suggesting stronger opaque priming in more experienced readers. For orthographic morphological priming, the interaction was significant in the opposite direction ($\beta = 117.69$, $p = 0.00029$) suggesting weaker, or more inhibitory, orthographic priming in more experienced readers.

8.2.3 Changing effects of OSC on priming with reading acquisition

For semantically transparent morphological prime-target pairs, the three-way interaction of word recognition accuracy, OSC, and priming was not significant ($\beta = -5.59$, $p > 0.8$). For semantically opaque morphological prime-target pairs, participants' word recognition accuracy did significantly moderate the effect of OSC on priming magnitude ($\beta = 726.95$, $p < 0.0001$). Orthographic priming also showed a change in the effect of OSC as word recognition accuracy increased, in the same direction as that for opaque priming

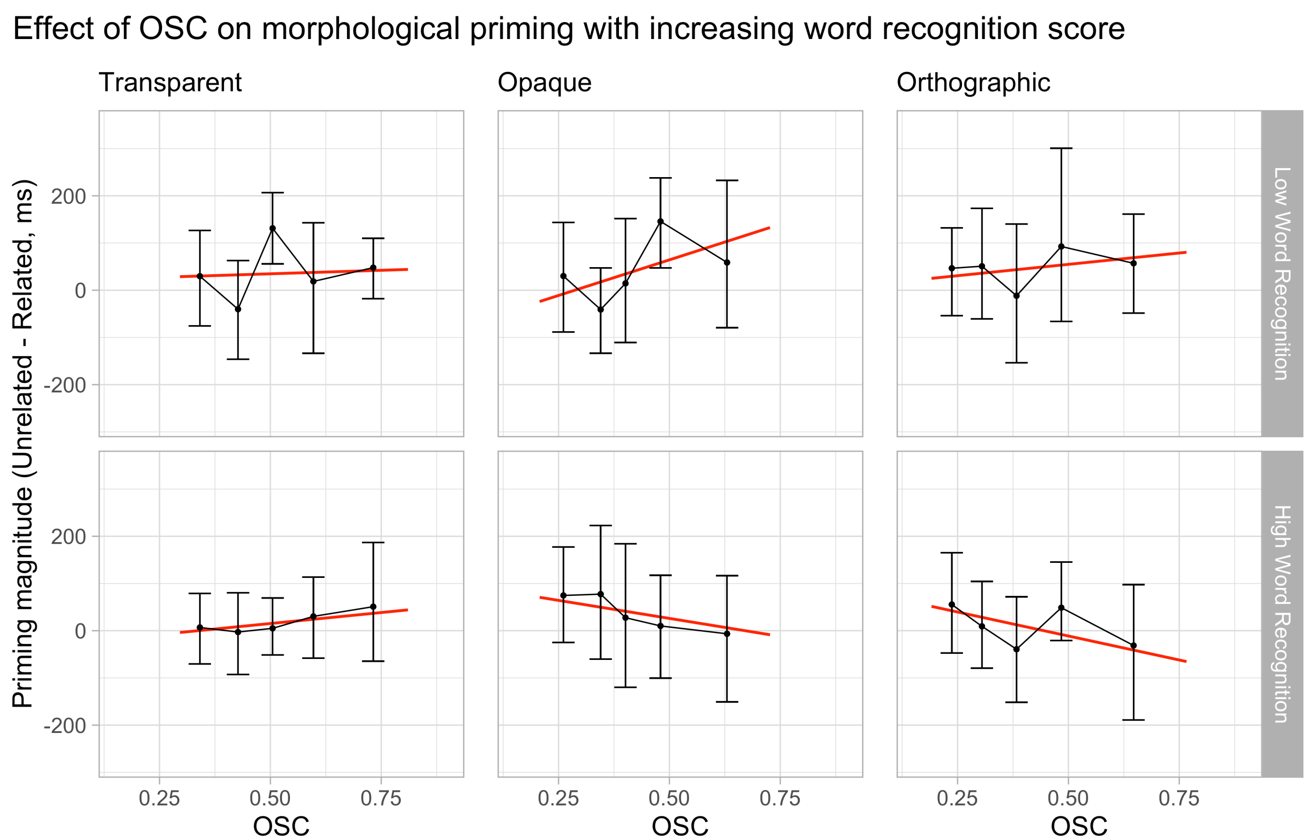

Figure 8.2: Priming magnitudes with increasing OSC, median-split by participant word recognition score. Each point represents mean priming magnitude by OSC quintile; error bars represent bootstrapped 95% confidence intervals. Red line shows line fit for by-item priming magnitudes.

($\beta = 577.46$, $p < 0.0001$).

To determine the direction of OSC effects on opaque priming for less and more experienced readers, the interaction of OSC and priming was examined separately for participants with word recognition accuracy at or below, versus above, the median (79%; see Figure 8.2). For orthographic priming, participants with lower word recognition accuracy showed a significant effect of OSC on priming such that priming magnitudes increased with greater OSC ($\beta = -119.77$, $p = 0.02$). A trend in the opposite direction was found for participants with higher word recognition accuracy ($\beta = 57.99$, $p = 0.09$). In the case of opaque priming, although neither interaction was significant, the strengthening effect of OSC on priming was greater in magnitude for participants with low word recognition ($\beta = -75.29$, $p = 0.28$) than those with high word recognition ($\beta = -27.87$, $p = 0.55$). This suggests that in both opaque and orthographic priming, the facilitation of stem OSC is stronger, or easier to detect, among less experienced readers.

8.2.4 Contrasting OSC effects on overt and masked priming in youngest age group

As only 4th and 5th graders were recruited for the overt condition of this study, word reading experience was not included as a moderator for models examining the implications of a longer prime duration. Instead, the three-way interaction of OSC, priming, and prime duration (masked or overt) was examined, and models were fit following the same procedure described above.

Prime duration significantly and positively moderated the effect of OSC on priming for all three conditions: transparent ($\beta = 177.469$, $p = 0.0012$), opaque ($\beta = 270.91$, 0.00185), and orthographic ($\beta = 510.35$, $p < 0.0001$). Looking at youngest-group participants receiving masked or overt priming separately, the effects of OSC on both transparent and opaque priming went from facilitatory for masked ($\beta = -158.11$, $p = 0.0018$ for transparent; $\beta = -108.32$, $p = 0.070$ for opaque) to neutral for overt ($\beta = 44.55$, $p > 0.4$ for transparent; $\beta = -20.39$, $p > 0.7$ for opaque). In the case of orthographic priming, the effect of OSC went from neutral when the prime was masked ($\beta = 12.25$, $p > 0.8$) to strongly inhibitory when the prime was overt ($\beta = 299.19$, $p < 0.0001$). Overall, in the case of transparent and opaque prime-target pairs, the effect of OSC was stronger and more facilitatory in masked relative to overt priming; for orthographic pairs, the effect of OSC was stronger and more inhibitory in overt relative to masked. (See Figure 8.3 for a visualization of these effects.)

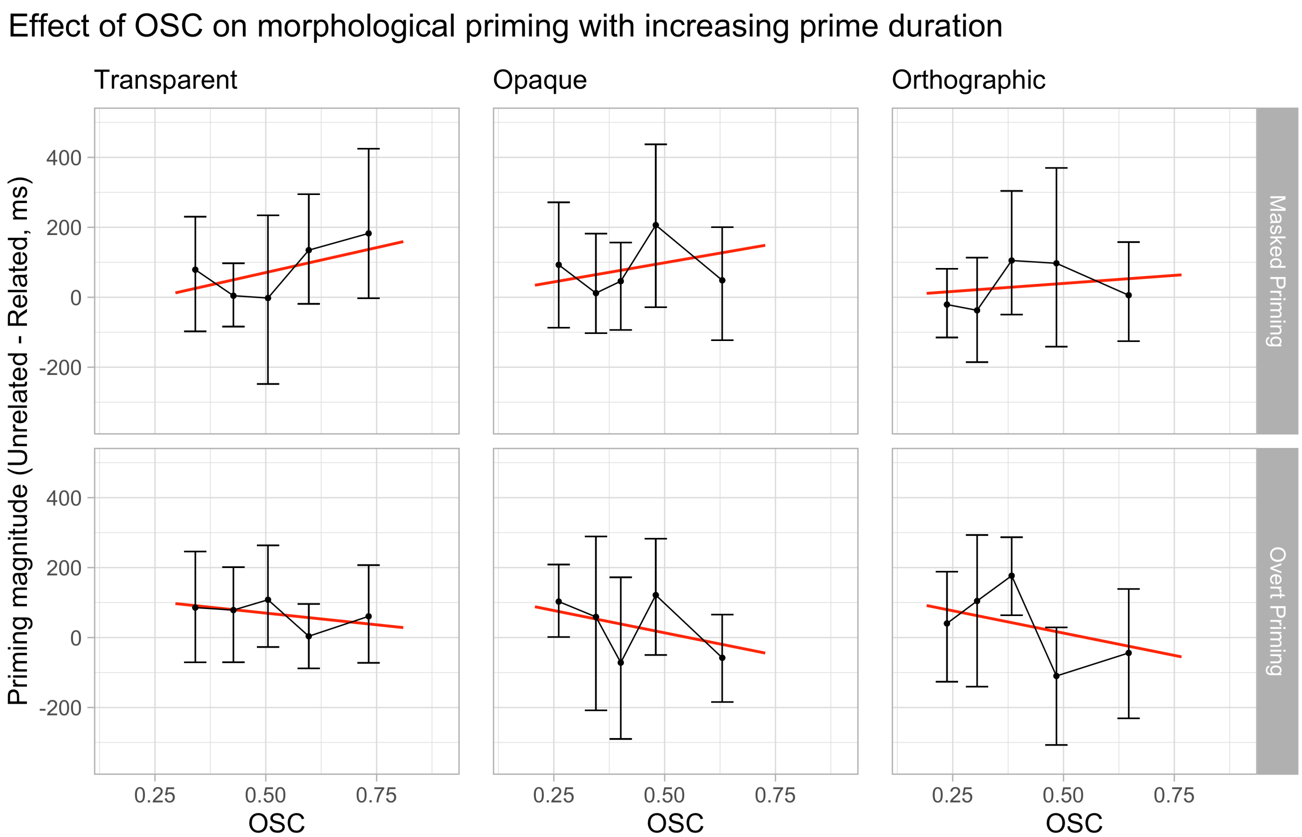

Figure 8.3: Priming magnitudes for masked versus overt priming with increasing OSC, for 4th and 5th grade participants. See Figure 8.2 for details of figure generation.

8.2.5 Investigating effects of morphological awareness

Performance on the three morphological awareness subtests administered at the end of the experiment were significantly correlated (all rs > 0.3, all ps < 0.001). Thus, accuracies on these subtests were combined for a cumulative metric of morphological awareness. Mean participant accuracy was 82.3%, and these scores were moderately correlated with participants' word recognition score ($r = 0.34$, $p = 0.0011$)

Analyses that included morphological awareness scores were conducted to answer two primary questions: First, does morphological awareness explain additional variance in morphological masked priming after word recognition has been accounted for? And second, does morphological awareness predict priming magnitudes differently in the context of overt priming relative to masked priming?

To address the first question, the residuals of the models fit to assess the effect of word recognition skill on overall priming in each condition (not including OSC as a predictor) were analyzed to examine whether the interaction of morphological awareness and priming explained additional variance. No significant effects of participants' morphological awareness performance on residualized priming magnitudes were found in any of the three priming conditions (all ps > 0.15).

To address the second question, the three-way interaction of morphological awareness, priming, and whether the prime was masked or overt was assessed for each priming condition. In all three conditions, the three-way interaction was significant such that better morphological awareness more strongly corresponded to greater priming in the overt priming context (for transparent priming: $\beta = -434.9$, $p < 0.0001$; for opaque priming: $\beta = -183.19$, $p = 0.00026$; for orthographic priming: $\beta = -258.22$, $p < 0.0001$). For contrast, with word recognition score in place of morphological awareness, there was no three-way interaction in the case of opaque priming ($p > 0.4$), while the interaction was negative for transparent priming ($\beta = -191.62$, $p < 0.0001$) but positive for orthographic priming ($\beta = 357.51$, $p < 0.0001$). Higher morphological awareness scores in young readers appear to more closely correspond to greater sensitivity to the presence of stems in overt primes than in masked primes, regardless of that prime's overall morphological structure.

8.3 DISCUSSION

As word characteristics (e.g., target and prime frequency) were not matched across conditions, the relative magnitudes of orthographic, opaque, and transparent priming cannot be compared with the current data. However, changes in magnitudes for each kind of priming with reading experience could be examined: as word recognition skill increased, transparent priming remained constant, opaque priming increased, and orthographic priming decreased (or rather became more inhibitory). These results align with those of Dawson et al. (2021), where opaque priming was found to significantly increase with word recognition experience relative to orthographic priming, but not relative to transparent priming. Their result could be attributed to an increase in opaque priming, a decrease in orthographic priming, or both, as found in the current study. This alignment of our results with the most robustly conducted prior study of morphological processing development in English gives us confidence in interpreting the rest of the findings presented.

For both semantically opaque morphological prime-target pairs and orthographic prime-target pairs, OSC was found to more strongly facilitate priming in less experienced readers than in more experienced readers. This result speaks to the primary objective of the current study, and aligns with our original predictions regarding the revised calculation of OSC: as a metric of how "meaningful" a particular letter string is across words, it should correspond to stronger and earlier-emerging sensitivity to the presence of morphological structure.

For participants with lower word recognition performance, the effect of OSC was detected only in cases where the word is not truly morphologically and semantically related.

It may be that the benefits of a high-OSC stem are overpowered when a more relevant, truly morphological relationship between prime and target is present. However, a significant positive effect of OSC on transparent priming *was* found in the analysis contrasting overt and masked primes, when only the youngest age group was included in analysis. A follow-up study with briefer prime presentation primes, or primes presented for the same duration but in an in-person, less variable context, might show OSC effects for transparent priming more consistently. The strong role of OSC in orthographic in addition to opaque priming also merits discussion; it appears that less experienced readers are sensitive to OSC in both cases, even though orthographic primes did not have an overall appearance of morphological structure. Given that OSC is calculated based on all words that contain the stem, regardless of their morphological structure, it is perhaps unsurprising that OSC might moderate orthographic priming as well as opaque. It is beyond the scope of the current study to determine whether OSC moderated opaque priming more strongly than orthographic priming (due to the lack of matched word characteristics across priming conditions), but this question may merit further investigation.

There are several potential explanations for the observed reduction of OSC's effect on opaque and orthographic priming in more experienced readers. Experienced readers process and recognize words faster: A 50-ms prime, which in young readers only allows access to more general characteristics of substrings independent of their current context, could allow for access to more context-sensitive information in older readers. This would explain both the decreasing role of orthographic-semantic consistency and the simultaneous strengthening of opaque priming and weakening of orthographic priming magnitudes found in this study as participants' word recognition skill increased. Alternatively, instead of sensitivity to OSC being simply easier to detect in less experienced readers, it could be that OSC is genuinely made use of more by these readers, whereas older readers have shifted away from OSC towards more informative or context-sensitive characteristics. Differentiating between the first explanation and this one would be difficult, but it could be potentially done with a more precisely implemented study, varying prime durations (e.g., 30, 50, and 70 ms) within multiple reading skill groups. Finally, it could simply be that, as readers recognize targets more quickly, there is less room for facilitation and thus effects are harder to detect. However, the overall magnitudes of facilitatory opaque priming did increase with word recognition skill, demonstrating that there was not simply a general ceiling effect with increasing skill.

The contrast of masked and overt priming for 4th and 5th grade participants demonstrated reduced facilitation of transparent and opaque morphological priming by OSC when the prime was overt. A weakened or nonexistent effect of OSC for longer prime durations is compatible with our previous characterization of the word processing time-course: Information associated with orthographic features and more general utility are dominant for briefly presented primes, whereas semantic and context-sensitive information can be activated by a longer (and thus stronger) input. For orthographically related prime-target pairs, however, the effect of OSC went from neutral for masked priming to inhibitory for overt priming. It could be that consciously processed orthographic primes are more "distracting" if the shared stem has higher OSC, whereas for masked orthographic primes the orthographically-informed metric is helpful or neutral. Alternatively, it could be that higher-OSC targets are easier to inhibit than lower-OSC targets, as they are responded to more quickly in general. The latter explanation may be more likely, particularly if we believe that the impact of OSC as a characteristic of the prime-target relationship is overridden in an overt priming context.

A puzzling discrepancy exists between results from analyses of the effect of OSC with varying word recognition skill, and those for varying prime duration. The effect of OSC on opaque masked priming is facilitatory for both the younger age group ($\beta = -108.32$, $p = 0.070$) and the lower word recognition group ($\beta = -75.29$, $p = 0.28$). However, the effects of OSC on orthographic and transparent priming vary between these two, overlapping groups. The effect of OSC on orthographic masked priming is neutral for participants from the younger age group ($\beta = 12.25$, $p > 0.8$), but it is strongly facilitatory for participants from the lower word recognition group ($\beta = -119.77$, $p = 0.02$). Additionally, opaque and orthographic priming behave more similarly with respect to OSC effects in the first analysis, and opaque and transparent priming behave more similarly in the second. We attribute these differences to the fact that, although the youngest age group and the lower word recognition group are overlapping, they are not the same (see Table 8.1). Switching from analyzing the lower word recognition group to the younger age group entails a loss of 18 lower-performing participants from the middle and older age groups, and the addition of 5 higher-performing participants from the lower age group. The change in OSC effects for orthographic priming could be due to a loss of power, and the change in transparent priming could be due to a slight overall increase in word recognition, although we do not have the data to support this speculation directly.

Finally, we had not expected to find that morphological awareness would explain additional variance in masked priming magnitudes after accounting for word recognition skill, and our findings confirmed this prediction. However, morphological awareness scores did better predict overt priming magnitudes relative to masked priming magnitudes, across all three priming conditions. This contrast aligns with the expectation that morphological awareness would play a greater role in how readers respond to primes that can be processed consciously, although the fact that the effect was found in all three priming conditions as opposed to just the transparent condition was unexpected. Apparently, participants with better morphological awareness were more generally sensitive to the presence of the stem in the overt prime, regardless of whether it was semantically transparent, semantically opaque, or followed by a non-affix ending. Perhaps this finding is a symptom of the young age of participants included in this contrast: a different pattern of effects may have been found if more experienced readers had also been presented with overt primes.

The primary goals of this study were (1) to further illuminate how morphological processing emerges with reading experience, (2) to investigate whether OSC predicts early emergence of morphological priming effects, and (3) to investigate how masked priming effects differ from overt priming effects in less experienced readers. Our results support those of Dawson et al. (2021) in showing a strengthening of opaque priming and a weakening of orthographic priming with reading experience. The finding that OSC more strongly facilitates opaque and orthographic priming for less experienced readers suggests that early sensitivity to morphological structure may be driven by higher-OSC stems. However, this was not unique to seemingly morphologically structured (i.e., opaque) primes, but occurred for orthographic primes as well. The emergence of sensitivity to opaque priming distinct from orthographic priming appears to coincide with a diminishing importance of OSC, indicating increased sensitivity to more wholistic morphological cues with reading experience. Regarding the differences between masked and overt effects in the youngest age group, the failure of OSC to predict transparent and opaque overt priming aligns with our conceptualization of OSC as characteristic with more bearing on early word processing. Although some of these effects are weak or variable, the overall pattern of results enriches our understanding of how morphological processing mechanisms are acquired by contextualizing the role of OSC within the acquisition of sensitivity to morphological structure.

The current study has limitations, and thus results must be interpreted with caution. Most notably, the unmoderated and remote nature of data collection resulted in the loss of a great deal of data, and stimulus presentation was less precise than would be expected in an in-person context. The unmoderated context also required the use of a less conventional approach to measuring word recognition skill (accuracy in a lexical decision task, as opposed to accuracy in reading words aloud as in Morris, 2013), although we feel that the use of words spanning a range of reading levels made our alternative a relatively sound approximation. Were developmental participants more easily accessible at the time the study was conducted, a robust assessment of word recognition skill would have been administered, and participants would have been assigned to conditions based on their reading skill as opposed to their age (this was only a concern for the assignment to the youngest group to overt or masked priming, as manipulations were otherwise within-participant). Similarly, the morphological awareness measure was also shortened and likely less precise than the full one conducted in Nagy et al. (2006). Finally, the study is cross-sectional, and thus we are not tracking the change in a certain set of individuals' word recognition processing with more experience, but rather comparing individuals with presumably more and less experience (as determined by their word recognition score). Given prior evidence for individual variability in morphological processing effects (e.g., Andrews & Lo, 2013) and the generally variable nature of these effects (as evidenced by the difference in OSC effects on transparent and orthographic priming for the youngest age group relative to the group with lower word recognition skill), a longitudinal variation of the current study may be worth conducting.

Individuals in the midst of early reading acquisition are forming morphological sensitivity to more general paradigms (e.g., the addition of -MENT to a verb) and particular morphological families (e.g., -TRUST-) at the same time. However, skilled readers can also acquire new sensitivities when they learn new words and integrate this knowledge into their already developed word processing mechanisms. For example, high-schoolers studying for the SAT may encounter several previously unknown words sharing the same Greek or Latin root (such as ANTHRO- or -MORPH-). The effect of OSC on experienced readers' acquisition of sensitivity to novel morphological families is discussed and explored in the next chapter.

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